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Department of Mechanical and Mechatronics
Engineering

Design and Testing of a Coaxial Propellant Valve

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Summary

This report documents the design process, testing outcomes, and integration of the main oxidizer propellant valve, also called the injector valve, for Waterloo Rocketry's 2023 Rocket. The injector valve has the following purpose: to open when remotely commanded to allow propellant to flow into the engine to enable combustion, and to stay closed otherwise.

This injector valve is a clean-sheet, student researched and developed (SRAD) design that addresses many of the deficiencies of the injector valve assembly used on the previous rocket in 2022. The valve is a custom-machined assembly that uses a flow path coaxial with its pneumatic actuator to allow for dramatic space and mass savings, while allowing for higher mass flow rates compared to the previous design and alternative designs. Custom valve seats were designed and machined out of glass-reinforced PTFE, and a magnetic sensing solution was developed to allow operators to remotely sense the physical state of the valve. The use of flange ends allows for more compact and easier integration and assembly with the rest of the Rocket.

Several designs of valves, including commercial off the shelf (COTS) designs were considered, but ultimately the coaxial design was selected based on performance, integration, cost, and learning opportunity reasons. Waterloo Rocketry takes on many advanced technology development projects to produce in-house designs, this valve being an example. Although many of these projects may have higher development and schedule risk when compared to using an existing commercial design, they provide Team members with great learning opportunities and improves the performance of the rocket by allowing for designs tailored to the Team's application, while reducing cost and procurement time.

1 Introduction

Since 2018, the rockets flown by the Team have used a family of hybrid propellant engine developed for a capstone project for Waterloo Rocketry, nicknamed ‘Kismet’[3]. Traditional combustion rocket engines typically use either a solid motor, where the fuel is premixed and cast into a solid, or a liquid engine, where both fuel and oxidizers are fluids that are injected into the combustion chamber. In contrast, a hybrid rocket engine, one of the propellants is a fluid and the other is a solid that is cast into the combustion chamber. The Kismet Engine uses a fluid oxidizer nitrous oxide and a solid fuel HTPB. The nitrous is injected into the combustion chamber using the vapor pressure of the nitrous oxide in the propellant tank.

In the 2022 design cycle, the Rocket’s feed line from the bottom of the oxidizer tank to the inlet port of the engine consisted of an approximately 1 ft long section of plumbing that included a 1/2” valve, a short length of 1/2” hose, and various other 1/2” fittings. This was found to be severely restricting the flow fluid from the oxidizer tank to the engine, so the decision was made to increase the feed line diameter of the rocket to a nominal size of 1” and to reduce the length of the assembly as well. The existing injector valve assembly also had issues that stem from its size, mass, and difficulty of assembly. Thus a new larger injector valve assembly design that allows for increased propellant flow rate to the engine and that addresses the issues of the previous injector valve assembly is desired for the 2023 Rocket.

2 Design Constraints and Criteria

The summarized design constraints and criteria for this valve are given in Table 1 and Table 2.

Table 1: Injector valve design constraints

Constraint	Description
Safety factor on yield	$SF_y = 2.0$
Materials	Valve must use materials that are readily available (such as AL6061-T6, mild steels, SS304), with short lead times (<1 week), and that can be machined at the E5 or E3 machine shop.
Default state	The valve must fail closed if pneumatic pressure or electrical power is lost.
Size	The valve assembly must fit inside the rocket's airframe, with enough available room around it to fit other components.
Competition	The valve must meet all competition rules and requirements of the 2023 Spaceport America Cup competition.

Table 2: Injector valve design criteria

Criteria	Description
Operating pressures	Maximum Allowable Working Pressure (MAWP): 1000 psi (69 bar)
Oxidizer compatibility	All materials used must be chemically compatible with nitrous oxide.
Operating temperature	The range of expected temperatures is -20°C to 40°C . This is to cover the range of operating conditions from engine testing in the winter at Waterloo to launch conditions in the New Mexico desert in June.
Mass flow rate	Must support a mass flow rate of 2.85 kg/s of nitrous oxide.
State feedback	There must be a way get direct feedback to determine the state of the valve (open or closed).
Pneumatic pressure	The actuator should be designed with a working pneumatic pressure of 100 psi (MAWP of the pneumatics system is 145 psi).
Mass	The mass of the assembly should reasonably be minimized, but mass-optimization features should be balanced against manufacturability.

3 Valve Selection

Several potential valve and actuator combinations considered for this new valve are summarized in Table 3.

Table 3: Possible valve and actuator combinations for the new injector valve

Valve type and Actuator	Description
Ball valve, piston with lever-arm	COTS 1" ball valve with an air piston, closest in design to the 2022 injector valve assemblyFigure 55
Ball valve, custom double piston rack-pinion	COTS 1" ball valve with a custom quarter-turn, double rack and pinion actuator
Poppet valve, air piston	Custom designed poppet valve with air piston for actuation
Coaxial valve, integrated piston	Custom designed coaxial flow valve with air piston integrated into the valve body

The decision was made to use pneumatics because it provides a lot of actuation force for the amount of space the gas takes up in the rocket, and to avoid the need to add heavy electric motors that could actuate this large 1" valve when under up to 1000 psi. Furthermore, the choice to use pneumatics can take advantage of the existing development work that was already done on the 2022 Rocket on-board pneumatics system, avoiding the need to develop, source, and integrate new control electronics for other types of actuators such as an electric motor.

3.1 Ball valve with lever-arm

Preliminary sizing of the ball valve and pneumatic piston solution showed that the required assembly would be too large and inconvenient to fit inside the airframe, as shown in Figure 1.

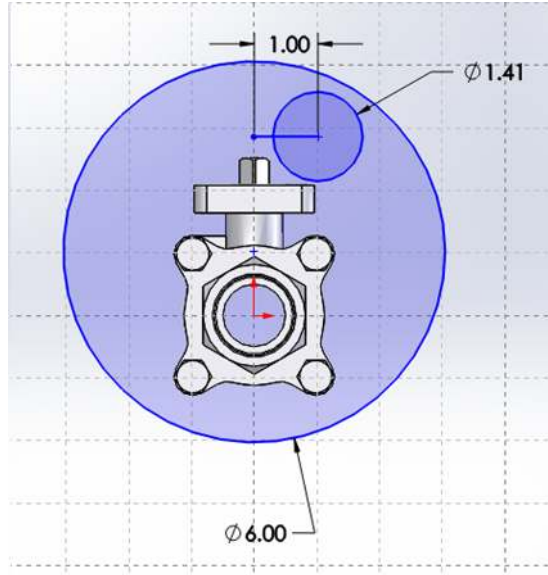


Figure 1: SolidWorks sketch of ball valve with required actuator inside 6" airframe

The required torque to turn the 1" 3-piece Assured Automation ball valve at the design pressure is 130 in·lbf[4, p. 6]. Assured Automation recommends a safety factor of 1.2 on top of this value[4, p. 6], giving 156 in·lbf required torque. Using a range of actuator arm lengths and the design 100 psi pneumatic pressure, a range of piston diameters can be determined. Equation 2 gives the required piston bore diameter for a given torque T_{req} , lever arm radius R , and pneumatic pressure P . A diagram of the piston and lever scenario is shown in Figure 2

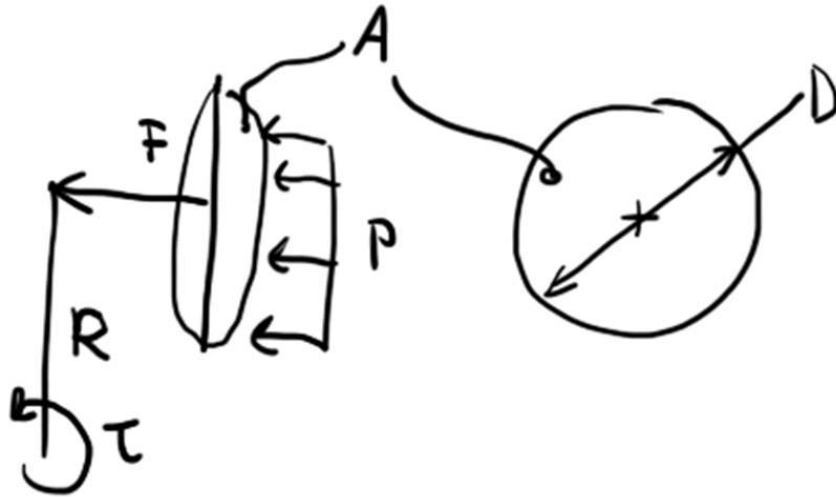


Figure 2: Schematic of pneumatic piston with lever

$$T_{\text{req}} = R \cdot F = R \cdot \frac{\pi}{4} D^2 P \quad (1)$$

$$\Rightarrow D = \sqrt{\frac{4T_{\text{req}}}{\pi R P}} \quad (2)$$

The required piston bore diameters for sample lever arm lengths are summarized in Table 4.

Table 4: Range of lever arm lengths with required piston bore diameters

Lever arm length [in]	Piston force required [lbf]	Piston bore diameter required [in]
0.5	312	1.993
1.0	156	1.409
1.5	104	1.150
2.0	78	0.996

3.2 Ball valve with double rack and pinion

The next option considered was the double rack and pinion actuator, which uses two opposing air pistons with rack gears to turn a pinion that is coupled to the stem of the valve as shown in Figure 3.

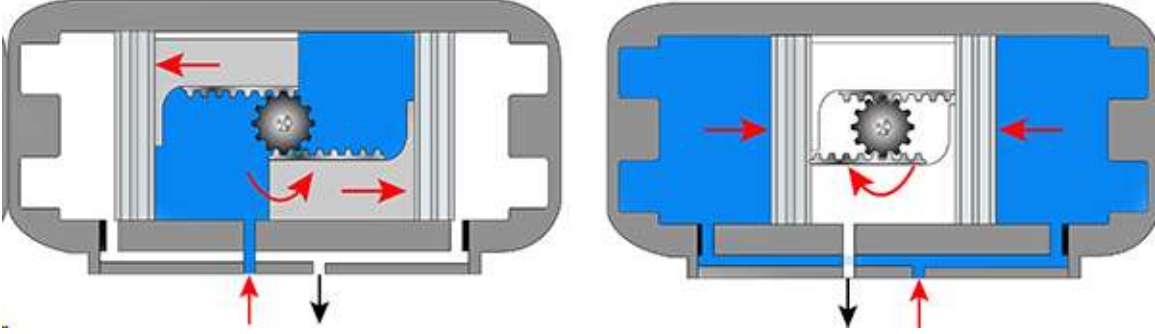


Figure 3: Pneumatic quarter-turn double rack and pinion actuator

The advantage of this approach is that it allows for a lot of torque to be generated in a relatively compact package due to both the double piston and the usage of gears. Unfortunately, COTS actuators of this design are too big to fit inside the airframe. For example, the Assured Automation CSR052, which is the smallest model that provides

adequate actuation torque at 100 psi, has a size of 5.79 in x 2.56 in x 3.62 in[5, p. 9], which does not fit inside the airframe when put on top of the 1" valve.

Another option explored was developing a custom quarter-turn actuator which uses the nitrous oxide propellant itself as a working fluid to actuate the piston. This has several advantages, including offering much higher actuation pressures of about 850 psi, the vapor pressure of nitrous oxide at 21°C. This helps reduce the size of the actuator by allowing more force to be produced. Using nitrous as a working fluid could also remove the need for the pneumatics system, eliminating a complex system that takes up space, mass, and that has the possibility of leaking and risking the mission.

Ultimately however, this option was not pursued because of the difficulty in developing a custom quarter-turn actuator that would be able to use nitrous oxide. Furthermore, this actuator would itself need a pilot nitrous valve. Commercial nitrous valve that could act as a pilot for this actuator were determined to be too heavy and drew too much power.

3.3 Poppet valve

The poppet valve consists of a flow path that turns 90 degrees and is controlled using a pneumatic piston that pushes a stem to create a seal between the stem tip and the valve seat. A schematic of a poppet valve is shown in Figure 4

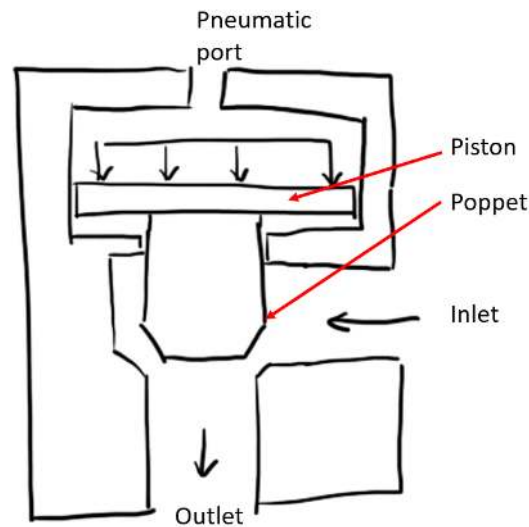


Figure 4: Poppet valve sketch

Although the design is not too complex, the main disadvantage of this valve design is the additional integration complexity and flow losses associated with the turns in flow path

inherent with this design. A sketch of how this valve can integrate inside the rocket is shown in Figure 5.

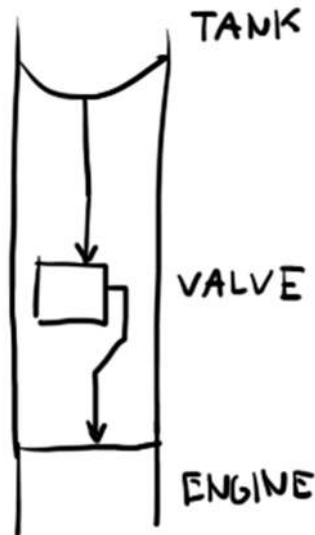


Figure 5: Sketch of potential poppet valve integration in rocket

The 90 degree turns and bent flow path would cause loss in flow performance of the nitrous and increased difficulty with integration inside the rocket.

3.4 Coaxial valve

The coaxial valve (also known as the sleeve valve in the aerospace industry) is a valve concept that uses a sliding hollow cylinder to seat against an annular valve seat. The flow travels in through the sleeve and then around the valve seat. The sleeve is actuated using a pneumatic piston, which is situated around the sleeve. The name of the valve comes from how the actuator and flow path are ‘coaxial’ with each other. A cross section of a commercial coaxial valve is shown in Figure 6.

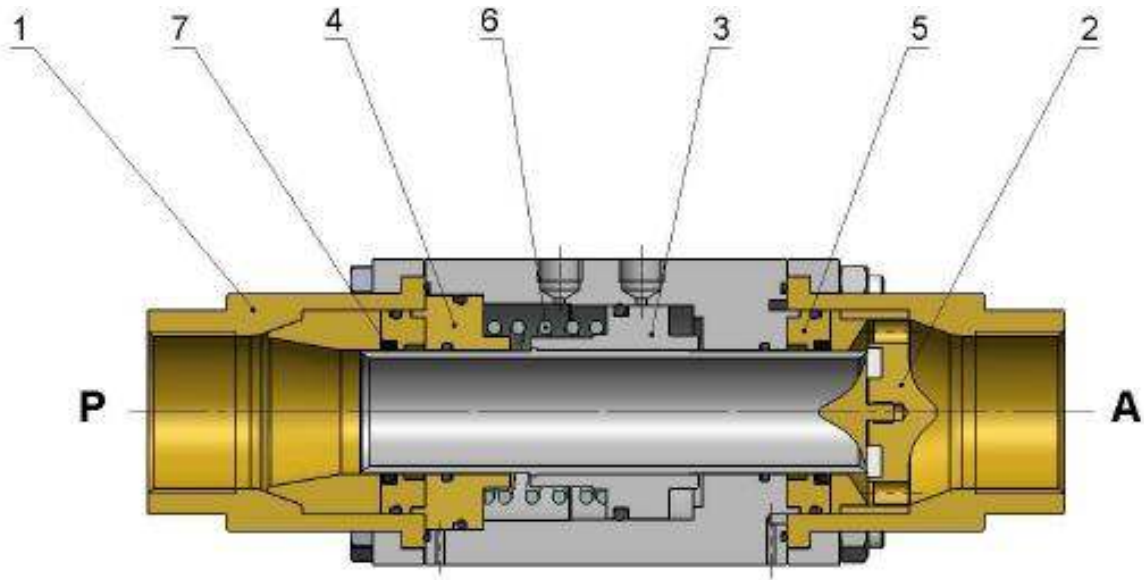


Figure 6: Cross section of a coaxial valve[1]

The advantage of this valve concept is that it is an inherently very compact and spatially-efficient design. Furthermore, the in-line geometry of the valve ports means that it can integrate easily in the center of the rocket, without needing to make any 90 degree bends or needing to offset the valve radially to fit a large actuator.

The main disadvantage comes from its level of development risk. The Team has never developed a fluid system assembly of this mechanical complexity before, and so the development risk associated with this option is higher. For example, one of the risks associated with this design is that it is uncertain how well a custom-designed and machined valve seat can seal against the sleeve, given the Team's lack of experience with this type of sealing and with machining thermoplastic materials such as PTFE for valve seats.

Overall, the pros and cons of each valve option is summarized in Table 5, with an item marked “-” indicating a con and an item marked “+” indicating a pro. The criteria evaluated for each valve include performance, cost, integration, and development risk.

Table 5: Pros and cons of each valve design

Criteria	Valve and actuator combination			
	Ball valve, piston with lever-arm	Ball valve, custom double piston rack-pinion	Poppet valve, air piston	Coaxial valve, integrated piston
Performance	+ Best propellant flow path – High air consumption – High mass	+ Best propellant flow path – Moderate air consumption – High mass	– Multiple bends in flow path needed + Low air consumption + Lower mass	+ minor detour in flow path + Low air consumption + Lower mass
Cost	– 1” Assured Automation ball valve ~\$198[6] – Solenoid valve ~\$45[7]	– 1” Assured Automation ball valve ~\$198[6] – Solenoid valve ~\$45[7] + Rack and pinion mechanism machined in-house	+ Entire valve machined in-house – Solenoid valve ~\$45[7]	+ Entire valve machined in-house – Many seals, hardware, components to buy – Solenoid valve ~\$45[7]
Integration	– Very tight integration, valve must be offset	– Very tight integration, valve must be offset	– Multiple bends needed to connect to valve	+ Easier integration since both flow path and actuator are coaxial
Development risk	+ Low development risk	– Moderate development risk (Rack and pinion). Riskier if pursuing nitrous actuation	– Moderate development risk (Poppet seat/tip)	– Moderate development risk (PTFE seat)

Based on primarily the development scope and integration concerns of the ball valve and poppet valve options, the decision was made after consultation with other stakeholders on the Team to pursue the coaxial valve option. A hand sketch of the coaxial valve design is shown in Figure 7.

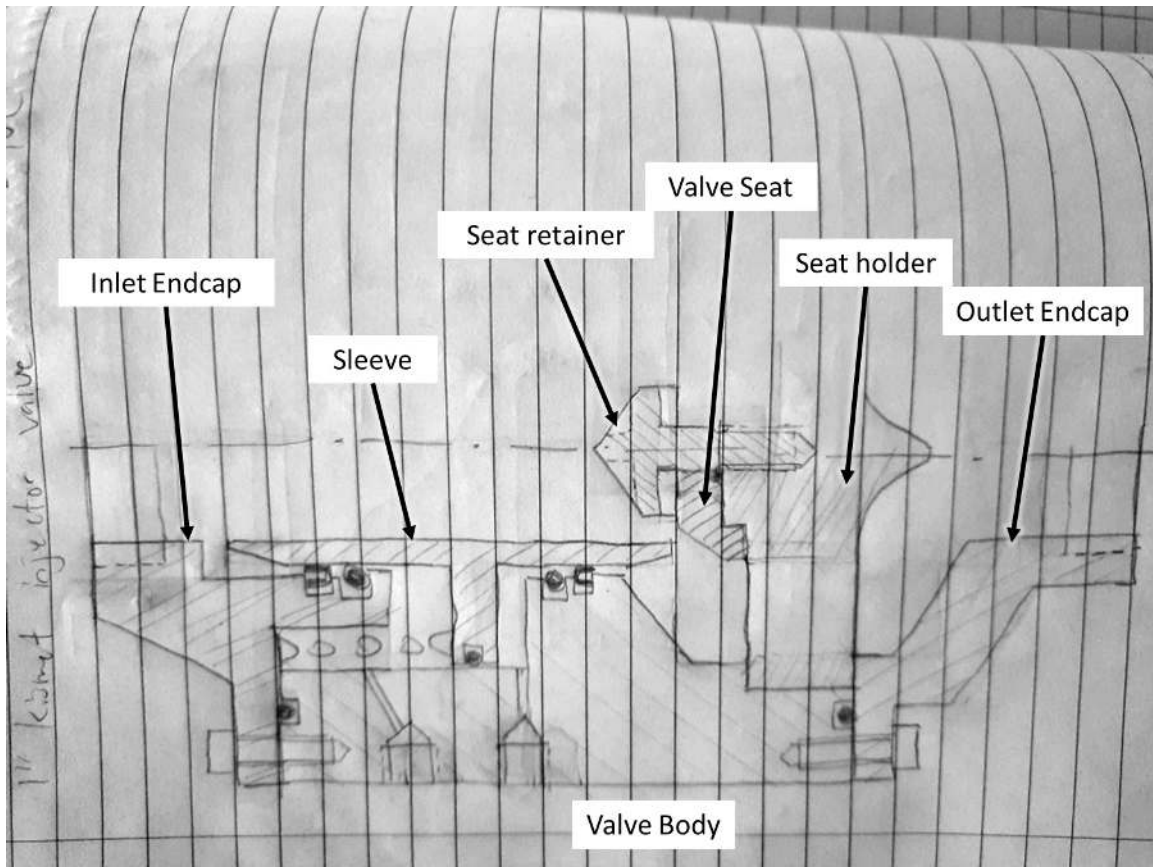


Figure 7: Hand sketch of the coaxial valve design

4 Flow Path Sizing

The 0.87" ID of the flow path was chosen due to match a commonly available commercial size of 1.000" OD tubing, which has .049" wall. This size of ID was also chosen to avoid losses from expansion or contraction of the flow when adapting the valve to the 1" tubing. A sketch of the flow path around the valve seat is shown in Figure 8.

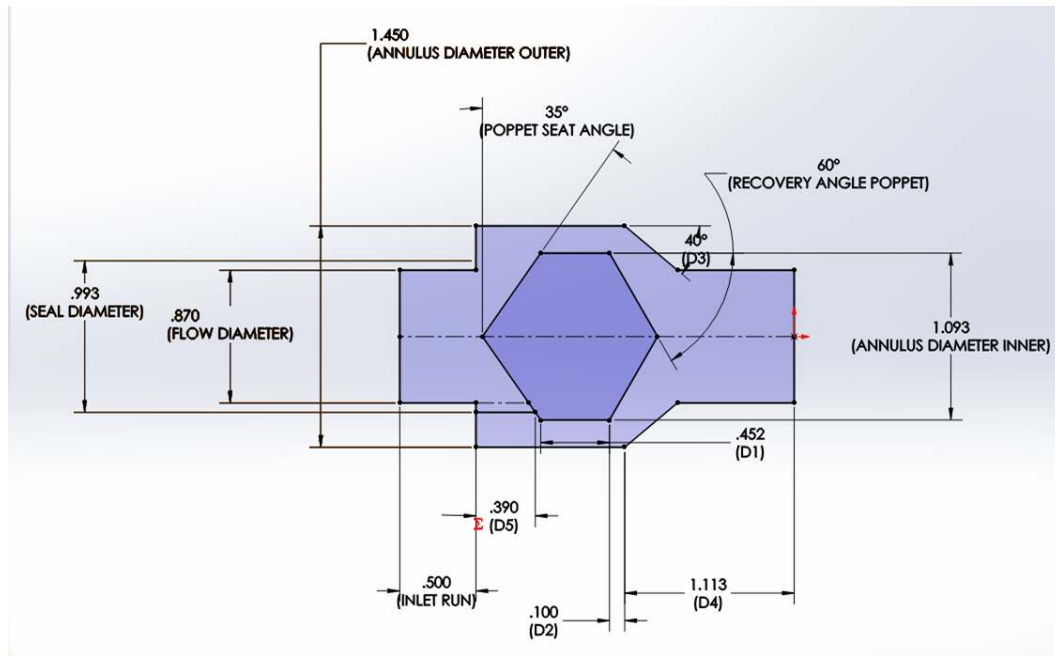


Figure 8: SolidWorks sketch of the flow path

4.1 Seat design

The seat angle is 35°. This angle was chosen considering the following factors:

- An acute angle is chosen to reduce the total length of the valve seat to reduce the valve length and the length of stock needed to machine the seat
- It is easier to machine an angle on the lathe when the angle is a multiple of 5 deg
- Literature that details the methodology of choosing a specific seat angle is difficult to find and a decision needed to be made to continue to move the project forward

The material of the valve seat is 25% glass-filled PTFE. This material was chosen because it is chemically compatible with nitrous oxide, stronger than non-reinforced PTFE, and is less expensive than specialized fluoropolymers such as PEEK or PCTFE. Based on the seat

geometry, a stroke length can be determined disregarding the seat angle effect by equating the area of the flow passage to the area of the cylindrical surface projected from the ID of the sleeve to the surface of the seat, as shown in Figure 9.

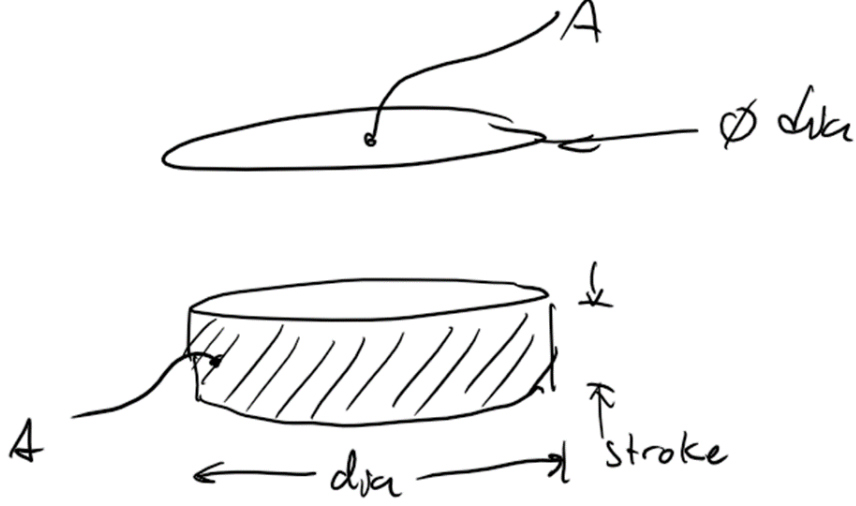


Figure 9: Flow passage between the valve seat and sleeve tip

Table 6: Stroke length calculations

Description	Value	Unit
ID diameter of flow passage	0.87	in
Area of flow passage	0.5945	in ²
Minimum stroke length	0.2175	in

The final stroke length that was used for the valve ended up being 0.39 in due to wanting more area when the flow diverts around the valve seat. The seat configuration for the first iteration of the valve is shown in Figure 10.

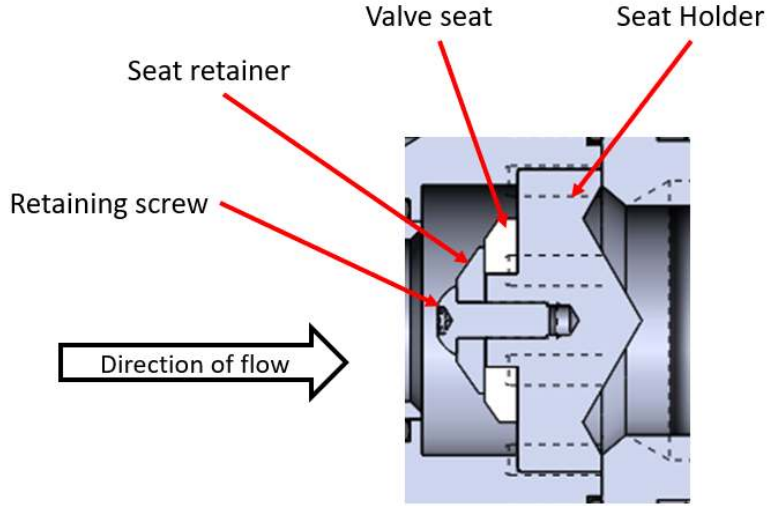


Figure 10: Seat configuration of prototype valve, with components labeled

The valve seat has a 0.500 in reamed bore in the center, which fits into a corresponding boss on the seat holder with a locating fit. The seat is then retained using a machined washer and a button head screw, which has a more contoured profile compared to a socket head screw or flathead. The initial design did not have a secondary seal for the seat nor a lip on the seat holder, which caused issues with seat deformation during testing. More detail can be found in the section Testing under ‘Valve seat radial deformation’ and ‘Valve seat axial deformation’.

4.2 Annulus sizing

After the flow passes around the PTFE seat, it must flow through an annular opening on the seat holder. Based on preliminary sizing of the valve seat in the seat holder, the annulus inner diameter was determined to be 1.093 in (equal to the seat OD). The outer diameter of the annulus is found by setting the area of the annulus equal to 1.2 times the flow area of the tube. This arbitrary factor of 1.2 was applied to the flow area was an initial guess to account for the size of radial spokes necessary to support the seat holder, which protrude into the flow area. The calculation of this annulus OD is summarized in Table 7.

$$A_{\text{annulus}} = \frac{\pi}{4} (d_o^2 - d_i^2) \quad (3)$$

$$d_o = \sqrt{\frac{4A}{\pi} + d_i^2} \quad (4)$$

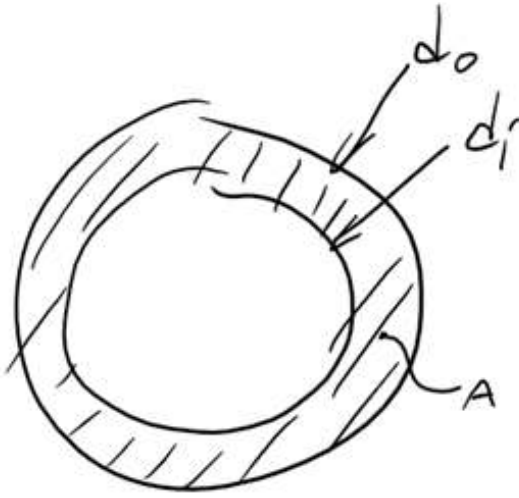


Figure 11: Annular flow cross section through seat holder

Table 7: Annulus diameter calculations

Description	Value	Unit
Area of flow passage	0.5945	in ²
Spoke Factor	1.2	
Area of annulus	0.7134	in ²
Annulus inner diameter	1.093	in
Annulus outer diameter	1.450	in

The SolidWorks screenshot below shows the seat holder after sizing the spokes, with the dimensions relevant to the flow area shown.

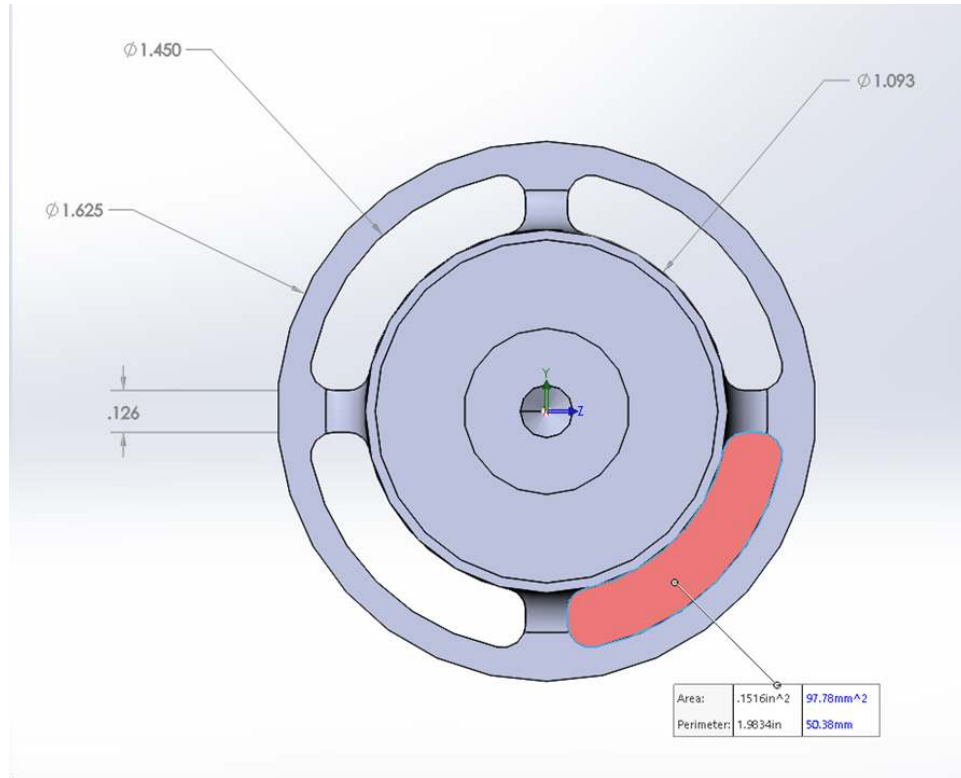


Figure 12: SolidWorks screenshot of seat holder, with annulus area highlighted

The total area of the flow passage through the slots is 0.6064in^2 . The seat holder was machined out of 6061-T6 aluminum stock.

5 Sleeve Sizing

The sleeve is the moving cylindrical piece that is actuated by a combination of spring force, pneumatic pressure, and imbalance of hydrostatic forces. The relevant dimensions for sizing the sleeve detailed in this section are indicated in Figure 13.

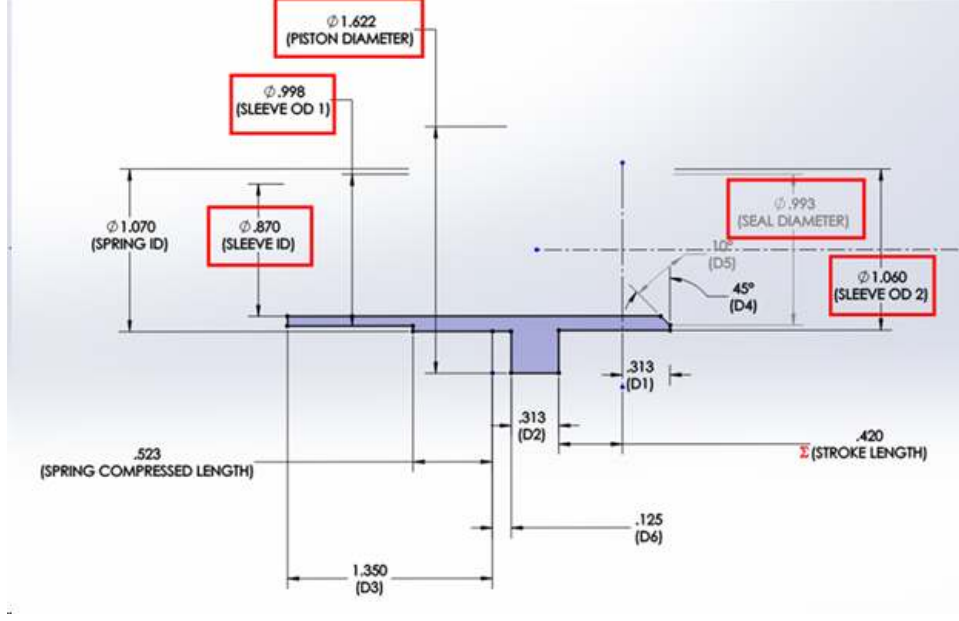


Figure 13: Half-cross section of the sleeve, SolidWorks sketch

The initial sleeve sizing is driven by the required thickness to contain the hydrostatic pressure of the fluid. The thin-wall assumption hoop stress equation (5) is used to calculate a wall thickness. The material chosen was aluminum 6061-T6 due to its availability and ease of machining.

$$\frac{S_y}{SF_y} = \frac{DP}{2t} \quad (5)$$

Table 8: Sleeve wall thickness calculations

Description	Value	Unit
Yield strength S_y	40000[8]	psi
Safety factor on yield SF_y	2	
Pressure P	1000	psi
Flow ID	0.87	in
Minimum wall thickness t_{min}	0.022	in

This gives an OD of $0.87 + 2(0.022) = 0.914$, which is $>20\times$ of t_{min} , justifying the thin-wall assumption. 0.021 in was determined to be too thin of a wall to machine, and deviations during machining could result in a lower wall thickness than planned for and

higher risk of failure. The wall thickness was increased to 0.044 in, which gives a very healthy margin of 102% on hoop stress.

$$MS_y = \frac{S_y}{S \cdot SF_y} - 1 \quad (6)$$

Table 9: Margins on yield for the sleeve

Description	Value	Unit
Yield stress S_y	40E3	psi
Safety factor on yield SF_y	2	
Pressure P	1000	psi
ID	0.87	in
Wall thickness t	0.044	in
Stress S	9886	psi
Margin on yield MS_y	102%	

Since the OD of the sleeve acts as the rod for a rod seal, the sleeve OD is selected as the next size up of dynamic rod seal dimensions from the Parker O-ring handbook. The dimensions for rod OD B, rod bore diameter D, rod gland groove diameter A-1, and groove width G are found from Parker Table 5-2, attached in the Appendix in Figure 51. The next size up of dimension for rod OD is 0.998 in. The dash size 214 dynamic rod gland is used.

The seating load required for the valve sleeve to form a good seal against the PTFE valve seat must be determined. To find this value, typically a loading rate (in force required per unit length of seal) specific to the seating geometry is used. For this specific sealing configuration, metal against PTFE poppet, a value of 4 lbf/in[9, p. 5] was found from a valve design that uses similar material for sealing (metal against PTFE) .

The next factor to consider is the imbalance in hydrostatic force on the sleeve when the valve is in a closed state. If both upstream and downstream side of the piston have the same area, but the sealing edge of the sleeve is along its ID, the piston will become unbalanced when the valve is in the closed position. This is because there is hydrostatic force on the annular area on the upstream face of the sleeve (top) but the corresponding face on the downstream side has no hydrostatic force acting on it, shown in Figure 14.

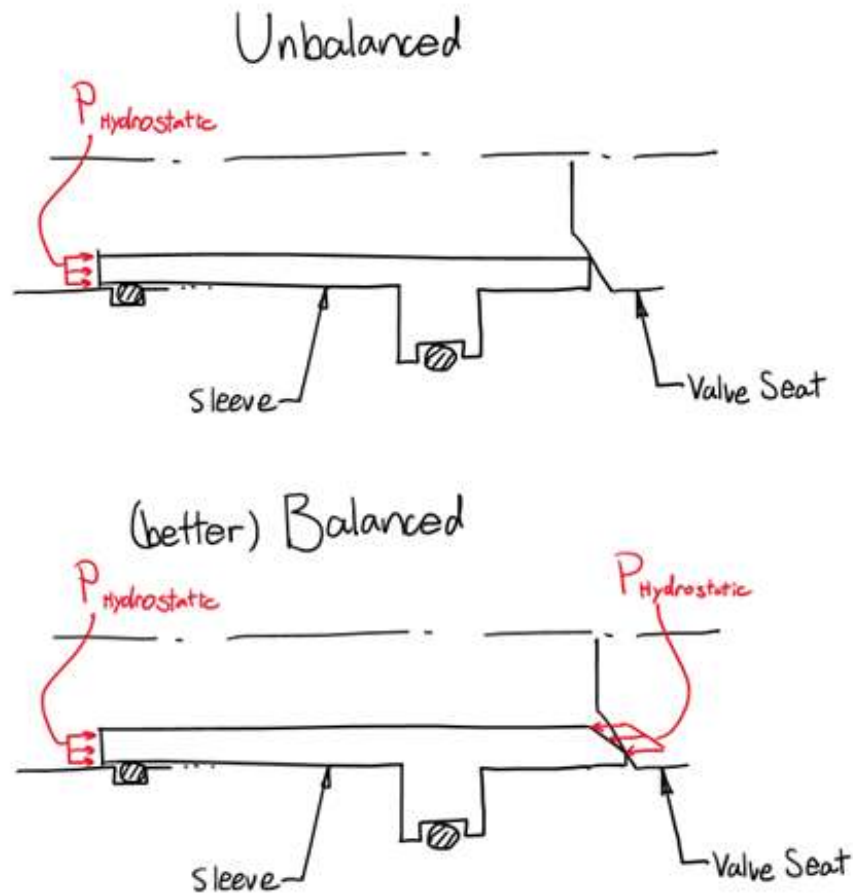


Figure 14: Unbalanced (top) and balanced (bottom) sleeve designs

This will cause excess seat load when the valve is in the closed position and increase the force requirements to open the valve, since the pneumatic piston will have to act against the hydrostatic force. To provide better balance on the sleeve, the diameter sealing edge of the sleeve is increased by introducing a chamfer on the downstream face. Some imbalance is still desired to help keep the valve naturally closed when under fluid pressure.

Since the area imbalance depends on the diameter of the sealing edge, an initial guess of 1.000 in for the new diameter of the sealing edge is made. The required seat load is calculated in Table 10.

Table 10: Required seat load calculation

Description	Value	Unit
Seat diameter	1	in
Seat circumference	3.142	in
Loading rate	4	lbf/in
Force required	12.57	lbf

Based on this seat load, the required area exposed to pressure on the downstream side of the sleeve can be determined. The net force on the sleeve is determined based on the MAWP of 1000 psi.

Table 11: Sealing diameter calculation

Description	Value	Unit
Force required	12.57	lbf
Pressure	1000	psi
Sleeve ID	0.87	in
Cylinder bore dia.	1.001	in
Force on upstream face	192.5	lbf
Area downstream required	0.1799	in ²
Seat diameter	0.993	in

This value is within 1% of the initial guess for the sealing edge diameter. To implement this seat diameter, a 45 deg chamfer is added to the inside edge of the downstream side of the sleeve. The chamfered edge will be forming the seal against the valve seat. Due to this chamfer, the end of the downstream side of the sleeve is now very thin, so the OD of the downstream side of the sleeve is increased according to the next size up of dynamic rod seal, dash size 215. The relevant dimensions are shown in Figure 15.

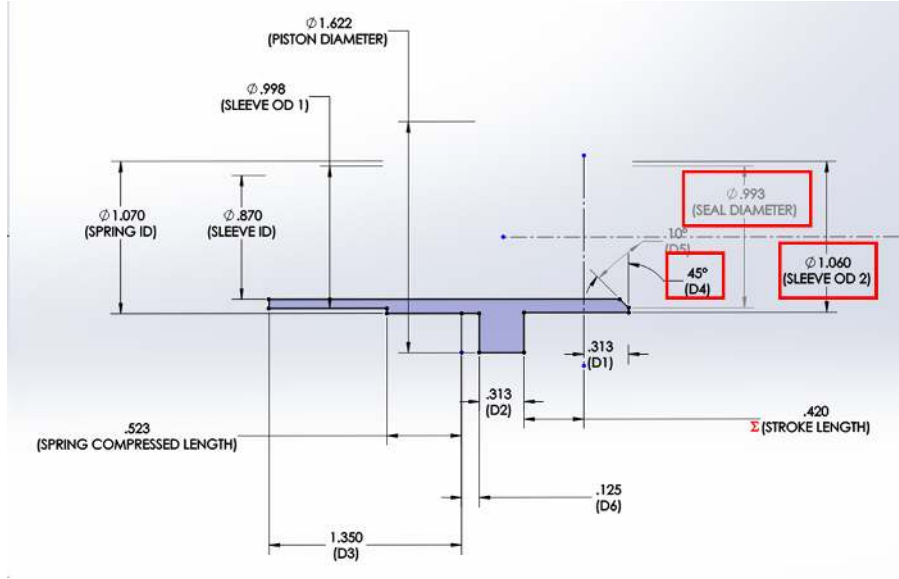


Figure 15: Sleeve dimensions, SolidWorks sketch

6 Actuator Sizing

The actuator is an annular piston on the sleeve that uses pneumatics to actuate at a design pressure of 100 psi. The methodology for actuator sizing consists of considering all possible states of the valve and the forces required to keep the valve in its current state or to transition to other states. The area of the air piston must be sized so that it has enough force to satisfy these 4 cases:

- Hold closed
- Actuate open
- Hold open
- Actuate closed

Additionally, a spring is used to provide a fail-safe bias to the valve, causing it to fail closed in case pneumatic pressure is lost. Furthermore, the imbalance of the hydrostatic force acting on the sleeve can also be used to aid in actuation.

6.1 Hold closed case

When the sleeve is closed, the actuator must provide enough force to adequately load the valve seat. The hydrostatic force also affects the force balance on the sleeve.

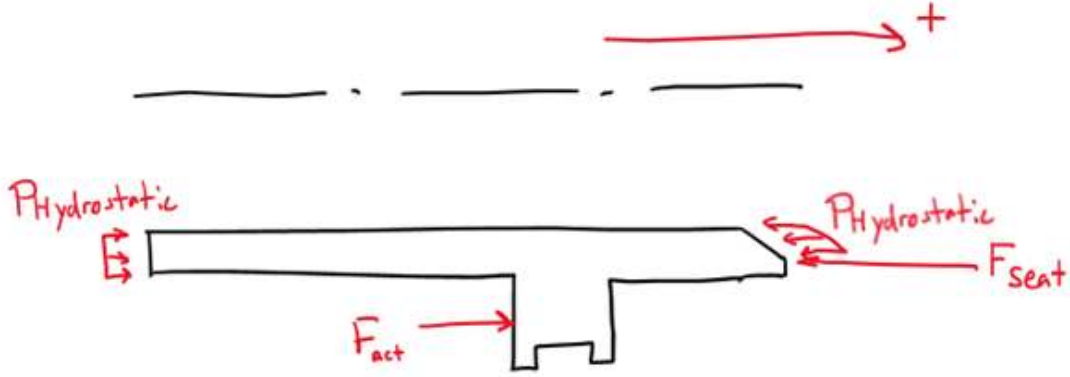


Figure 16: Free body diagram of sleeve, hold closed case

$$F_{\text{hydrostatic}} + F_{\text{act}} - F_{\text{seat}} = 0$$

$$F_{\text{seat}} = \text{Seat Diameter} \cdot \pi \cdot \text{Loading rate}$$

$$F_{\text{hydrostatic}} = P \cdot A_{\text{net}}$$

(positive area = upstream side, negative area = downstream side)

Seat Diameter = 0.993 in

Loading rate = 4 lbf/ in

$P = 1000$ psi

$$A_{\text{net}} = A_{\text{upstream}} - A_{\text{downstream}} = \frac{\pi}{4}(0.998^2 - 0.87^2) - \frac{\pi}{4}(0.993^2 - 0.87^2)$$

$$= 0.007819 \text{ in}^2$$

$$F_{\text{act}} = 4.66 \text{ lbf}$$

6.2 Actuating open case

In this case, the actuator force must be enough to start opening the valve from a closed state. The actuator must overcome the unbalanced hydrostatic force on the sleeve, the biasing spring, and the static friction of the three O-rings: upstream rod seal, downstream rod seal, and the actuator piston seal. Since friction of the actuator piston seal is determined by the size of the O-ring which depends on the actuator sizing to begin with, the friction

from the piston O-ring is instead approximated conservatively to be dash size 220 dynamic piston seal.

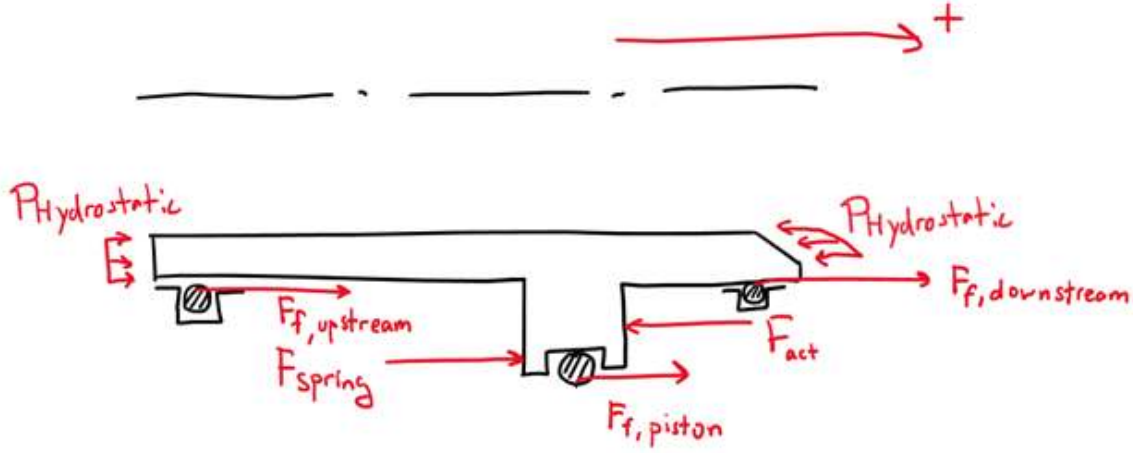


Figure 17: Free body diagram of sleeve, actuating open case

$$F_{\text{hydrostatic}} + F_f + F_{\text{spring}} - F_{\text{act}} = 0$$

$$F_f = F_{f,\text{upstream}} + F_{f,\text{downstream}} + F_{f,\text{piston}}$$

$$F_{\text{hydrostatic}} = P \cdot (A_{\text{upstream}} - A_{\text{downstream}})$$

When the valve is just opening, the hydrostatic pressure only acts to the extent of the sealing edge on the downstream side of the sleeve.

$$F_{\text{hydrostatic}} = 1000 \cdot \frac{\pi}{4} (.998^2 - .993^2) = 7.818639 \text{ lbf}$$

At the start of the stroke, the spring provides enough force to satisfy the seat load, which is 12.48 lbf. $F_{\text{spring}} = 12.48 \text{ lbf}$

The total O-ring friction consists of two components, friction due to seal compression and friction due to hydraulic pressure on the seal. They can be found by applying a coefficients of friction due to seal compression on the rubbing length of the seal, and by applying a coefficient of friction due to hydraulic pressure on the projected area of the seal gland [2, p. 5-7].

To determine the coefficient due to seal compression, the durometer and the compression (squeeze) of the O-ring must be found. The O-ring material selected is FKM (Viton) because it is chemically compatible with nitrous oxide. The durometer of FKM O-rings is Shore 75A, from McMaster P/N 9464K36. The seal compression for a radial seal can be found with the

following formula[2, p. 2-14]

$$\text{Seal compression} = \left[1 - \frac{\text{Total radial clearance}}{\text{O-ring width}} \right] \times 100\%$$

For the Parker dynamic seal gland designs of size -2xx, this compression value comes to around 12% to 13%. The coefficient of friction is found by interpolating between the 70A and 80A curves in Figure 18. This gives $f_c = 1.241 \text{ lbf/in.}$

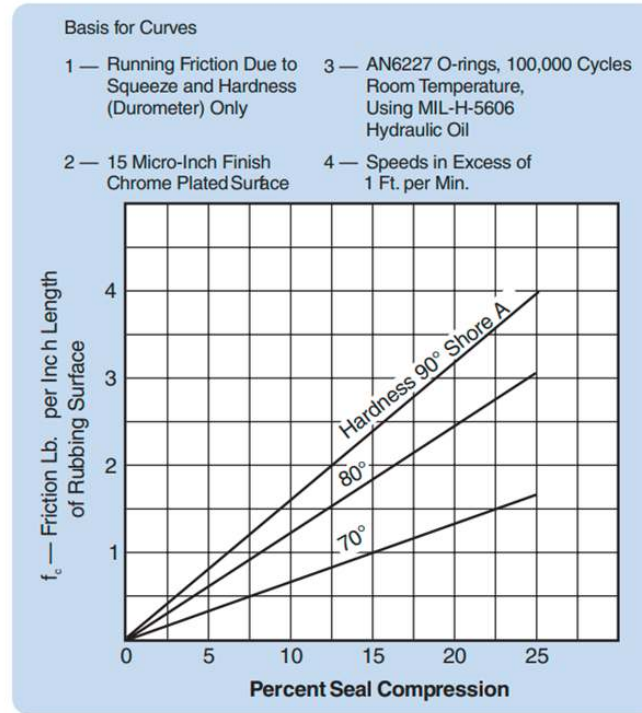


Figure 5-9: Friction Due to O-ring Compression

Figure 18: Coefficients of friction due to rubbing[2, p. 5-8]

The other coefficient of friction the seals experience depends on the fluid pressure and can be found using the chart in Figure 19. The upstream and downstream rod seals are exposed to up to 1000 psi and therefore experience about 37.5 lbf/in^2 . The piston seal is at pneumatic pressure, which is about 100 psi, and experiences about 15 lbf/in^2 .

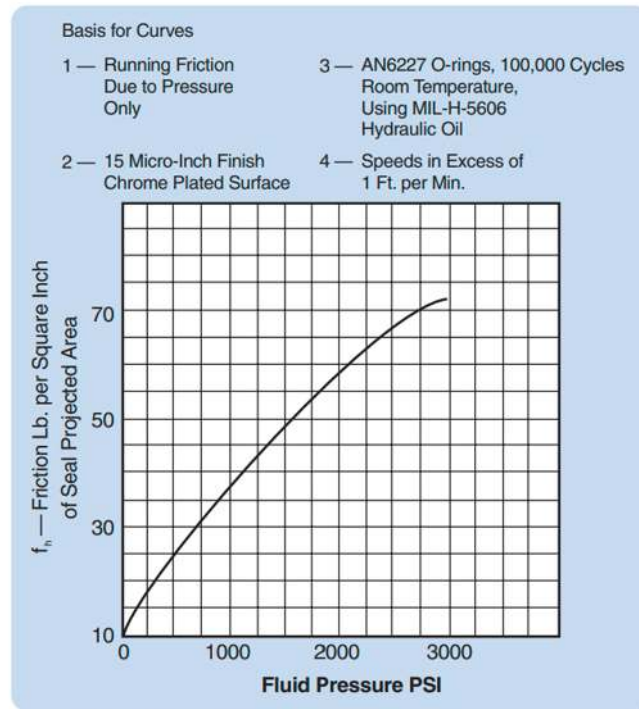


Figure 5-10: Friction Due to Fluid Pressure

Figure 19: Coefficients of friction due to fluid pressure[2, p. 5-8]

Table 12 summarizes the contribution to friction from each O-ring.

Table 12: O-ring friction calculations

Description	Upstream rod	Downstream rod	Actuator piston	Unit
Dash size	214	215	220	
ID	0.998	1.063	1.383	in
OD	1.240	1.302	1.625	in
Projected area (in ²)	0.4254	0.4439	0.5717	in ²
Rubbing length (in)	3.135	3.340	5.105	
Friction coeff. due to seal compression, 75A durometer	1.241	1.241	1.241	lbf/in
Rubbing friction (lbf)	3.89	4.14	6.34	lbf
Friction coeff. due to hydraulic pressure, 75A durometer	37.5	37.5	14	lbf/in ²
Hydraulic friction	15.95	16.65	8.00	lbf
Total friction	19.84	20.79	14.34	lbf
Sum	54.97			lbf

This gives a total friction of about 55 lbf acting in the positive direction. The force required at the start of the opening stroke is therefore

$$F_{\text{hydrostatic}} + F_f + F_{\text{spring}} - F_{\text{act}} = 0 \Rightarrow 7.82 + 55 + 12.48 - F_{\text{act}} = 0 \Rightarrow F_{\text{act}} = 75.3 \text{ lbf}$$

To ensure the force requirements are adequate to complete the stroke, the force exerted by the spring and the new hydrostatic force on the sleeve must be considered. At the end of stroke, hydrostatic pressure is acting on the full downstream face of the sleeve.

$$F_{\text{hydrostatic}} = 1000 \cdot \frac{\pi}{4} (.998^2 - 1.060^2) = -100.21 \text{ lbf}$$

The negative sign indicates the hydrostatic pressure now helps in actuation by pushing the sleeve away from the valve seat. Considering the spring at the end of the opening stroke, the spring will be at its most compressed state. Let $F_{\text{spring}} = 55 \text{ lbf}$, assuming the spring is sized such that it can overcome the O-ring friction.

$$F_{\text{hydrostatic}} + F_f + F_{\text{spring}} - F_{\text{act}} = 0 \quad (7)$$

$$\Rightarrow -100.2 + 55 + 55 - F_{\text{act}} = 0 \quad (8)$$

$$\Rightarrow F_{\text{act}} = 9.8 \text{ lbf} \quad (9)$$

Taking the greater of the two required forces, the required actuator force in this case is 75.3 lbf.

6.3 Hold open case

In this case, the actuator needs to resist the spring force pushing the valve towards closing.

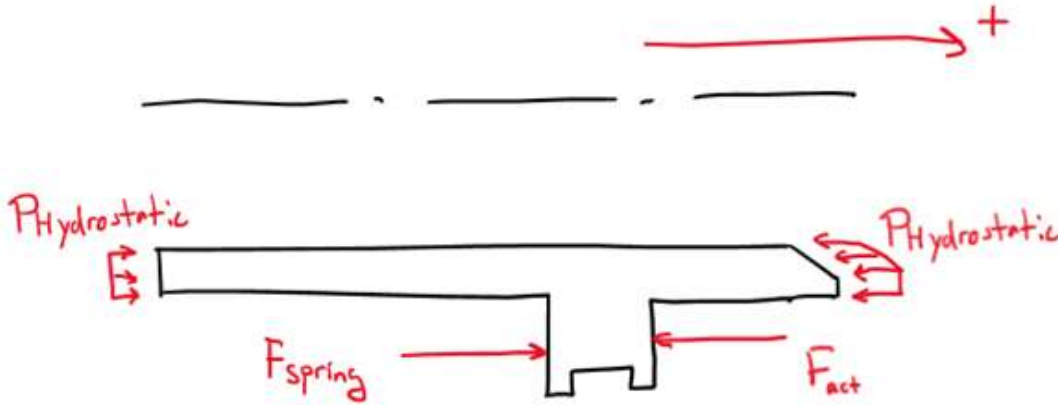


Figure 20: Free body diagram of sleeve, hold open case

$$F_{\text{hydrostatic}} + F_{\text{spring}} - F_{\text{act}} = 0$$

Due to the area imbalance, the hydrostatic pressure will provide 100.2 lbf in the negative direction, help to keep the valve open when at a pressure of 1000 psi. However, to ensure the valve can stay open even at low pressures, the force requirement in this case should be to resist the spring force. Again, the spring force when the sleeve is fully open is assumed to be 55 lbf.

$$F_{\text{act}} = 55 \text{ lbf}$$

6.4 Actuating closed case

When the valve is actuating closed, the forces acting on the sleeve are actuator force, O-ring friction, hydrostatic forces, and spring force.

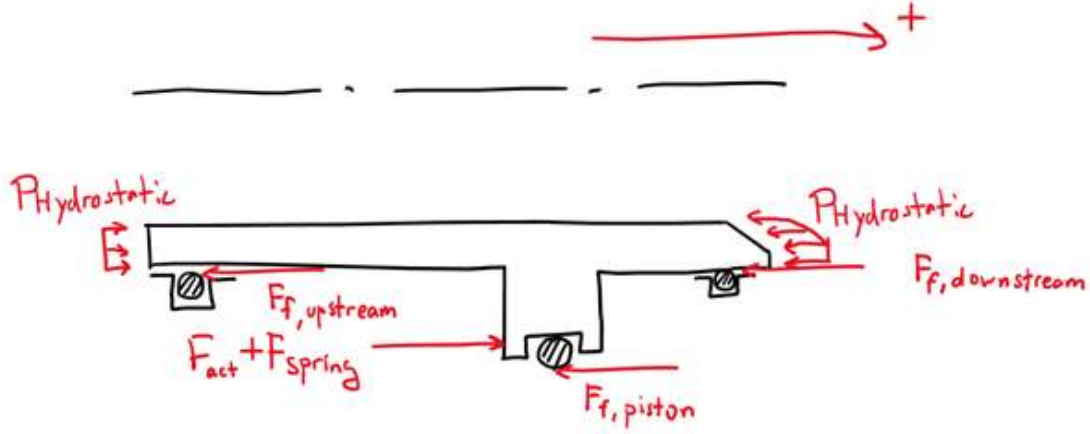


Figure 21: Free body diagram of sleeve, actuating closed case

$$F_{\text{hydrostatic}} - F_f + F_{\text{spring}} + F_{\text{act}} = 0 \quad (10)$$

$$F_f = 55 \text{ lbf} \quad (11)$$

From the previous case,

$$F_{\text{hydrostatic}} = -100.2 \text{ lbf} \quad (12)$$

$$F_{\text{spring}} = 55 \text{ lbf} \quad (13)$$

From Equation (10),

$$F_{\text{act}} = 100.2 \text{ lbf} \quad (14)$$

6.5 Piston OD and seal

The force requirements of the actuator are thus summarized in Table 13.

Table 13: Actuate closed force requirements

Description	Value	Unit
Hold closed	4.66	lbf
Actuate open	75.3	lbf
Hold open	55	lbf
Actuate closed	100.2	lbf

The driving cases for the piston sizing are the actuate open and actuate closed cases. Applying an arbitrary safety margin of 25% on the required actuation force, the required bore diameter for the upstream and downstream side of the piston can be determined. The calculations are summarized in Table 14.

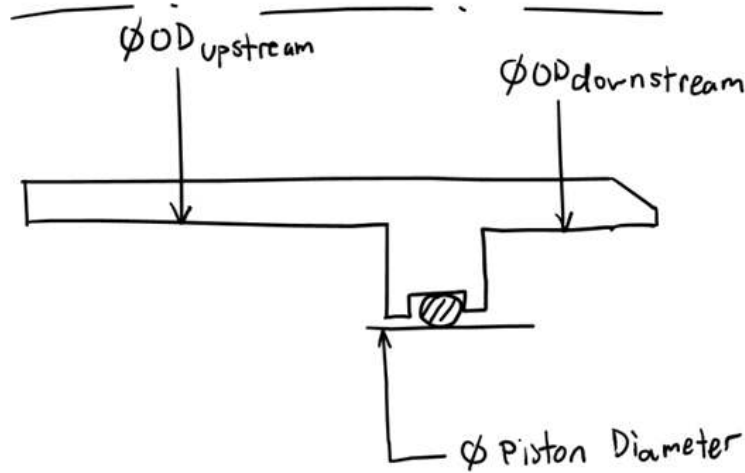


Figure 22: Piston and sleeve ODs

$$A_{\text{annulus}} = \frac{\pi}{4} (\text{OD}^2 - \text{ID}^2)$$

Table 14: Actuator piston sizing calculations

Description	Value	Unit
Actuate closed	100.2	lbf
Actuate open	75.3	lbf
Upstream OD	0.998	in
Downstream OD	1.06	in
Margin	25%	
Pneumatic pressure	100	psi
Upstream area required	1.2525	in ²
Downstream area required	0.9413	in ²
Upstream piston bore diameter	1.610	in
Downstream piston bore diameter	1.524	in

The minimum piston bore diameter is 1.610 in. The next size up of dynamic piston O-ring gland is size -220.

7 Spring sizing

From the sizing of the actuator and sleeve, the parameters that will drive spring sizing and selection are given below.

Table 15: Spring rate calculations

Description	Value	Unit
Stroke length, minimum	0.39	in
Extended force	12.48	lbf
Compressed force	55	lbf
Minimum ID	0.998	in
Spring rate, maximum	109.0	lbf/in

Since it was difficult to find a COTS spring that would satisfy all these exact requirements simultaneously, it was decided to fabricate a custom length spring using spring stock. The spring stock chosen was a 302 stainless steel compression spring McMaster P/N 9663K42, which satisfies the minimum ID condition.

The cut length required can be calculated using the spring constant information from McMaster.

$$\text{Rate of Cut Spring} = \frac{\text{Rate} \cdot \text{Length}}{\text{Cut Length}}$$

Table 16: Required cut spring length calculations

Description	Value	Unit
Spring rate required	109.0	lbf/in
length, uncut	20	in
Spring constant, uncut	5.6	lbf/in
length, cut	1.027	in
Force required, Valve open	55	lbf
Spring compressed length	0.523	in

This compressed length dimension is used to size the length of the upstream side of the valve and to size a shoulder on the sleeve that will act as the stop for the sleeve when actuating open.

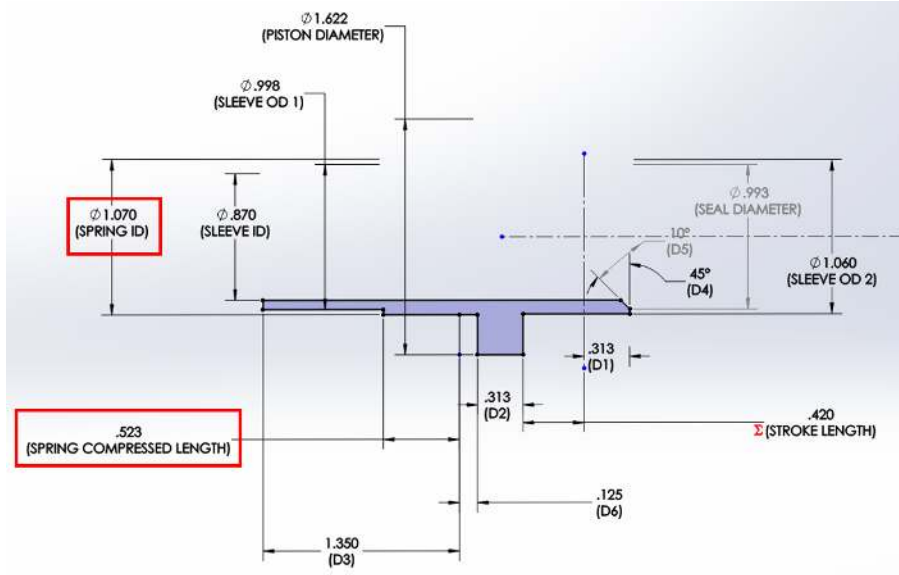


Figure 23: SolidWorks sketch of the sleeve, spring-related dimensions indicated

With the geometry of the piston determined, the pneumatic ports were added in, and two extra #4-40 blind mounting holes were added. Shown in Figure 24.

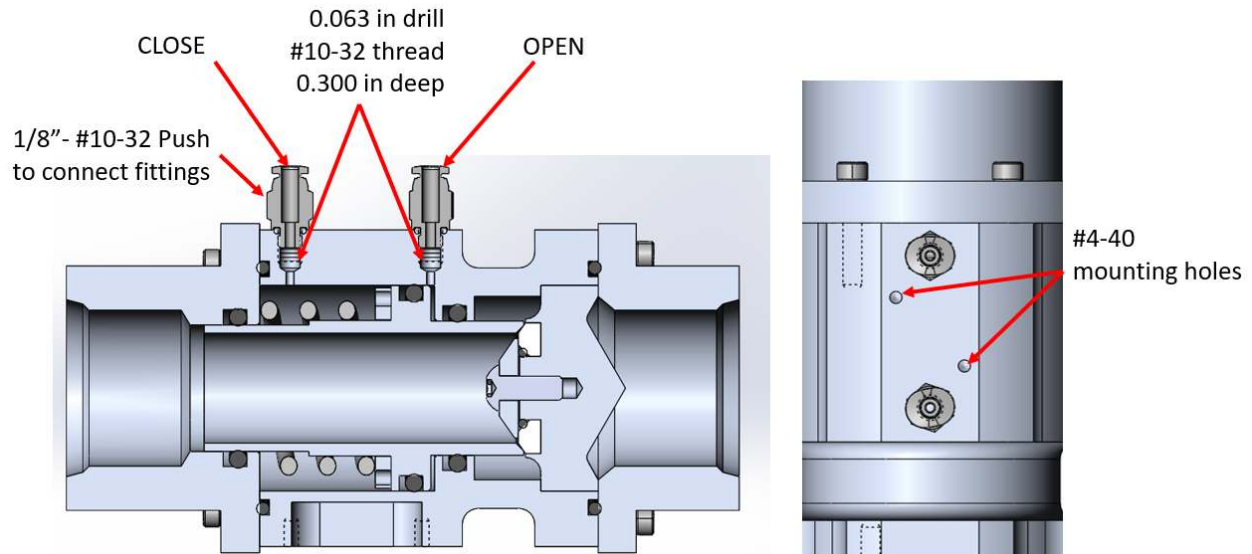


Figure 24: Valve pneumatic ports and mounting features

These holes allow for potentially mounting a pneumatic manifold or face-mount solenoid instead of push to connect fittings. This option would condense the entire system and save the space taken up by a pneumatic fittings and pneumatic lines.

Cut outs were also made on the valve body to reduce mass, shown in Figure 25. An estimated 124.8 g of mass was saved using 3 simple cutouts on the body.

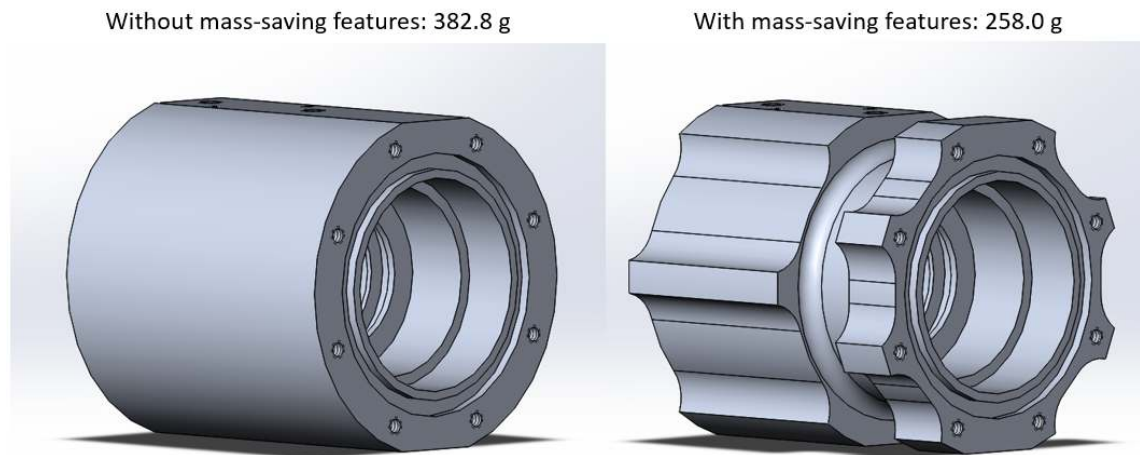


Figure 25: Comparison between valve body without (left) and with math saving features (right)

Table 17: Mass savings on valve body

Description	Value	Unit
Without lightening features	141.79	cm ³
With lightening features	95.57	cm ³
Density	2.7[8]	g/cm ³
Mass saving	124.8	g

8 Endcaps Sizing

The inlet and outlet endcaps retain the internal components of the valve and also provide a fluid interface through a threaded port or flange. The inlet and outlet endcaps are sealed with a static face seal and are fixed to the valve body with a bolted flange. To size the inlet and outlet endcaps, first the face seals are sized. The seal design is a standard FKM O-ring static face seal with the gland dimensions taken from the Parker O-ring handbook.

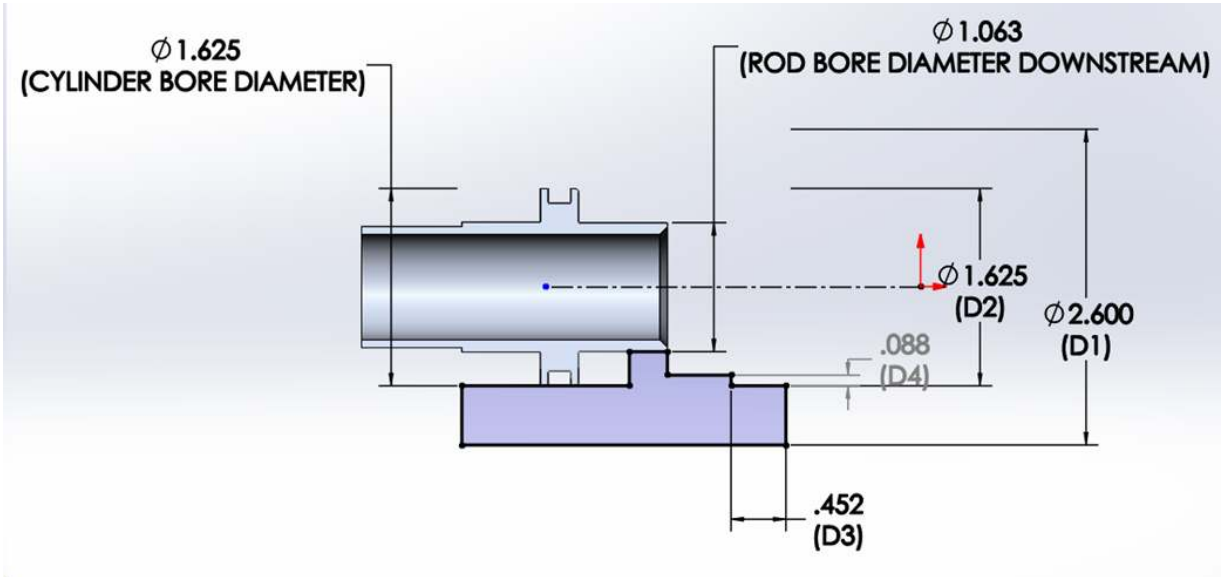


Figure 26: Bore diameters on the valve body

Based on the bore diameters in Figure 26, the minimum ID of seal must be 1.625 in, plus about 1/8" of retaining lip for the seal. To find the required OD, an additional 2 times 0.122 in for the width of the groove for a dash 1xx size of O-ring is added. This gives a minimum seal OD of 1.994 in. The smallest size of O-ring that meets this requirement is dash 133. This gives the following gland design based on static face seal design from Parker[2, p. 4-18].

This is shown in Figure 27. The same gland design is used on both inlet and outlet end flanges.

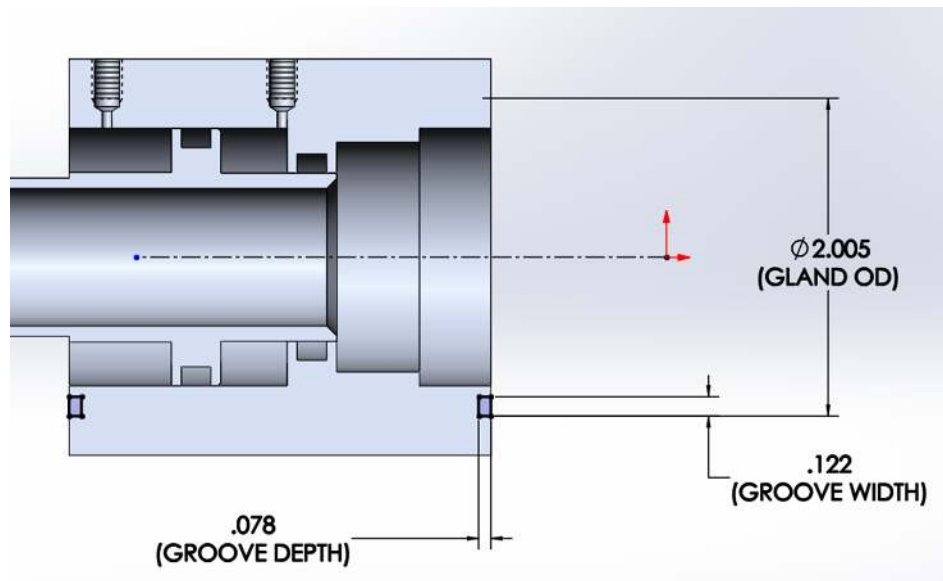


Figure 27: O-ring face seal gland on valve body

With the seal sized for both inlet and outlet flanges, the forces acting on them can be determined. For each of the inlet and outlet endcaps, there are two load cases to consider: one when the valve is actuated open and one when the valve is actuated closed. The possible loads on the endcaps include forces from:

- Nitrous oxide hydrostatic pressure
- Pneumatics
- The spring
- O-ring compression (squeeze)

O-ring compression, comes from the amount of force the flange needs to apply to the O-ring to compress the O-ring to form a good seal. According to the Parker O-ring handbook, the compression force is found by determining a compression load per linear inch of seal. This factor depends on the cross section of seal, its hardness, and the amount of compression. O-ring squeeze can be calculated with the following formula[2, p. 2-14]

$$\text{Squeeze \%} = \frac{\text{O-ring Width} - \text{Gland Depth}}{\text{O-ring Width}} \times 100\%$$

For this face seal, the squeeze is about 23%.

Table 18: O-ring compression calculations for face seal gland

Description	Value	Unit
-133 O-ring OD Width	0.103	in
Gland depth	0.079	in
Squeeze	23%	

The compression load factor is found in Parker Figure 2-5 (Appendix, Figure 49). In the 20% range, for a 75A durometer, the compression load factor is about 15 lbf/in. The compression force for the O-ring is therefore 94.48 lbf as summarized in Table 19.

Table 19: O-ring compression load required

Description	Value	Unit
-133 O-ring OD	2.005	in
Seal length	6.299	in
compression load factor	15	lbf/in
Compression load	94.48	lbf

8.1 Inlet endcap

Considering the closed case first, the loads are shown in Figure 28.

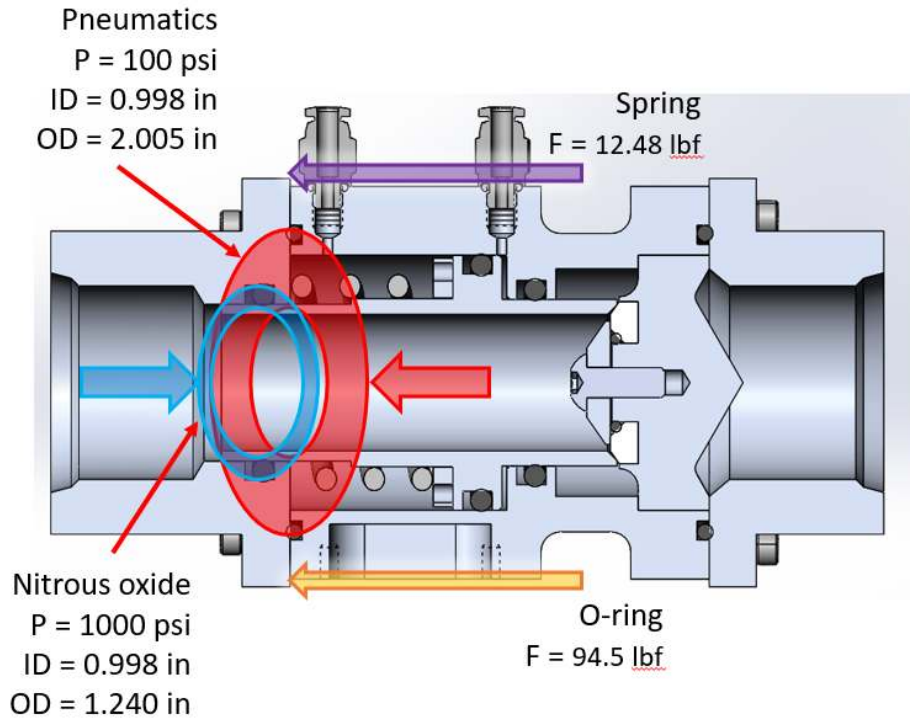


Figure 28: Areas and forces on inlet endcap, valve closed

The expected loads are summarized in Table 20 (positive is in the upstream direction).

Table 20: Forces on inlet end cap when valve is closed

Location	Area (in ²)	Pressure (psi)	Force (lbf)
Endcap Face	2.3751	100	237.5
Rod seal	0.4254	1000	-425.4
O-ring compression	-	-	94.5
Spring compression	-	-	12.48
Net Force			-80.88

In the worst case, there is no internal pressure to resist the force from pneumatics, so the load in this case should not include the force from the rod seal area. $F = 344.49$ lbf

Next is the opened case. Note the net load is shown on the sleeve, since part of the downstream area is balanced by the upstream area.

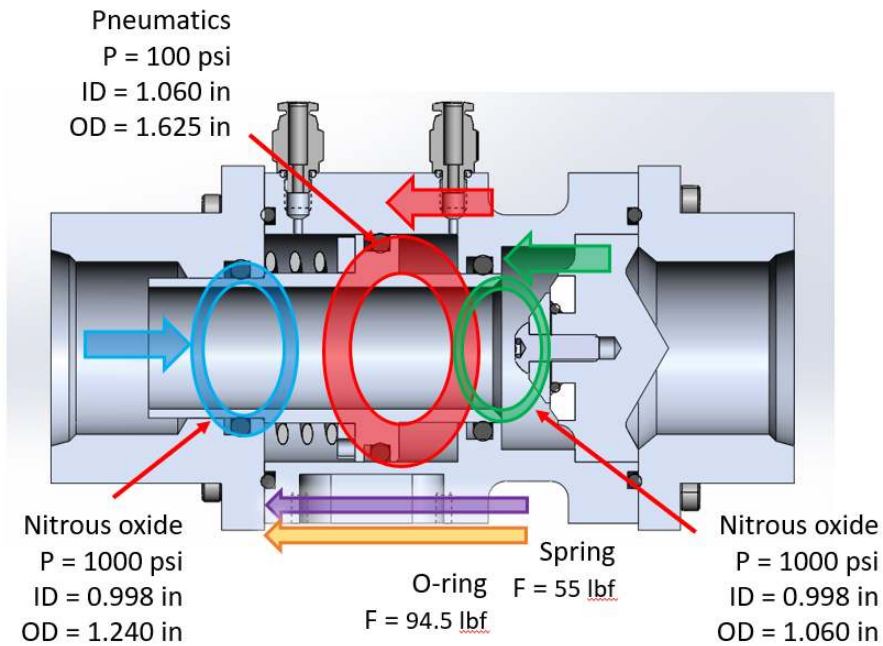


Figure 29: Areas and forces on inlet endcap, valve opened

The expected loads are summarized in Table 21 (positive is in the upstream direction).

Table 21: Forces on inlet end cap when valve is opened

Location	Area (in ²)	Pressure (psi)	Force (lbf)
Piston	1.1915	100	119.15
Rod seal	0.4254	1000	-425.37
Sleeve, net force	0.1002	1000	100.21
O-ring compression	-	-	94.5
Spring compression	-	-	55
Net Force			-56.51

In the case there is no internal pressure, the load in this case would not include the force from the rod seal area and the sleeve, which gives a net force of 268.65 lbf.

The design load for the inlet endcap is the greater of the two loads, 344.49 lbf. The bolts selected for this joint are black oxide alloy steel socket head cap screws per ASTM A574. These bolts were chosen for their good strength, availability, and compactness. To find the

tensile area of bolts required, the preload and proof strength must be found. The preload should be at least the load on the endcap to prevent significant forces from starting to load the bolt shank.

The preload is based on 75% of the proof load, or 85% of the yield strength of the bolt material if the proof strength is not known[11, p. 434]. The proof strength is 140 ksi[10]. With a safety factor of 2 on yield, the required bolt area can be calculated.

$$\frac{F}{A} = \frac{0.75S_p}{SF_y}$$

Table 22: Required bolt tensile area calculations, inlet endcap

Description	Value	Unit
Load	344.5	lbf
Safety factor on yield	2	lbf
Proof strength	140000	psi
Preload	75%	
Tensile area required	0.006562	in ²

The tensile area is used to find the number of bolts required for each size. Coarse threads were chosen to reduce the chance of cross-threading in the aluminum material. #4 screws were deemed too fragile and hard to work with due to their small size, so the next size up of screw was selected.

Table 23: Sizes and number of bolts required for inlet endcap flange

Bolt size	Tensile Stress Area[11, p. 405]	Bolts Required
#4 UNF	0.00661	1.0
#6 UNC	0.00909	0.7
#8 UNC	0.014	0.5
#10 UNC	0.0175	0.4

To distribute the clamping loads evenly around the bolt circle, six #6-32 screws were arranged into a bolt circle, shown in Figure 30.

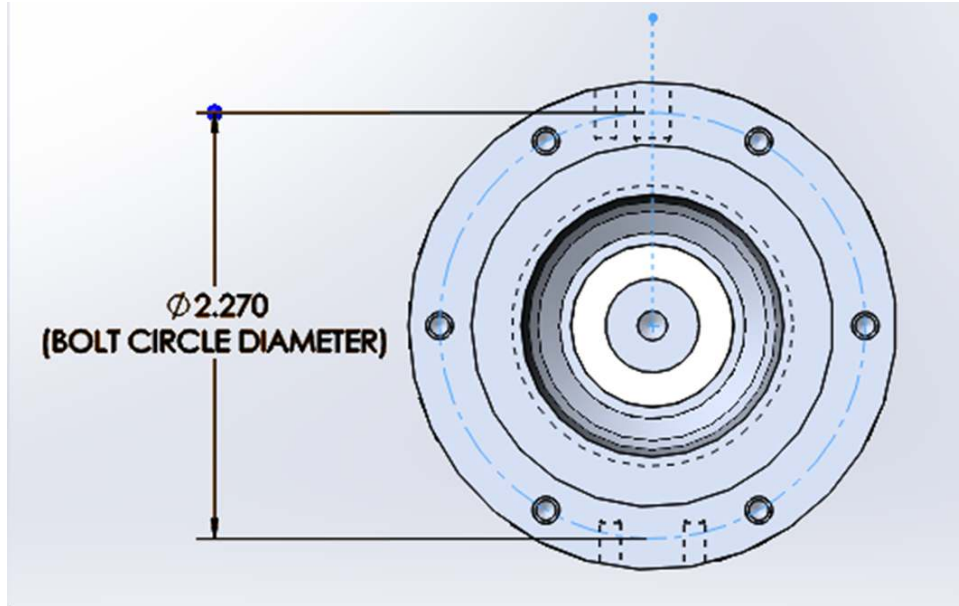


Figure 30: Bolt circle pattern on the inlet side of the valve body

8.2 Outlet endcap

The same methodology is used to size the bolt flanges on the outlet endcap. First considering the closed case, the pneumatic pressure, fluid pressure, and spring all push the sleeve in the downstream direction. This force is then transmitted through the seat holder and to the end cap. The loads are shown by the diagram in Figure 31.

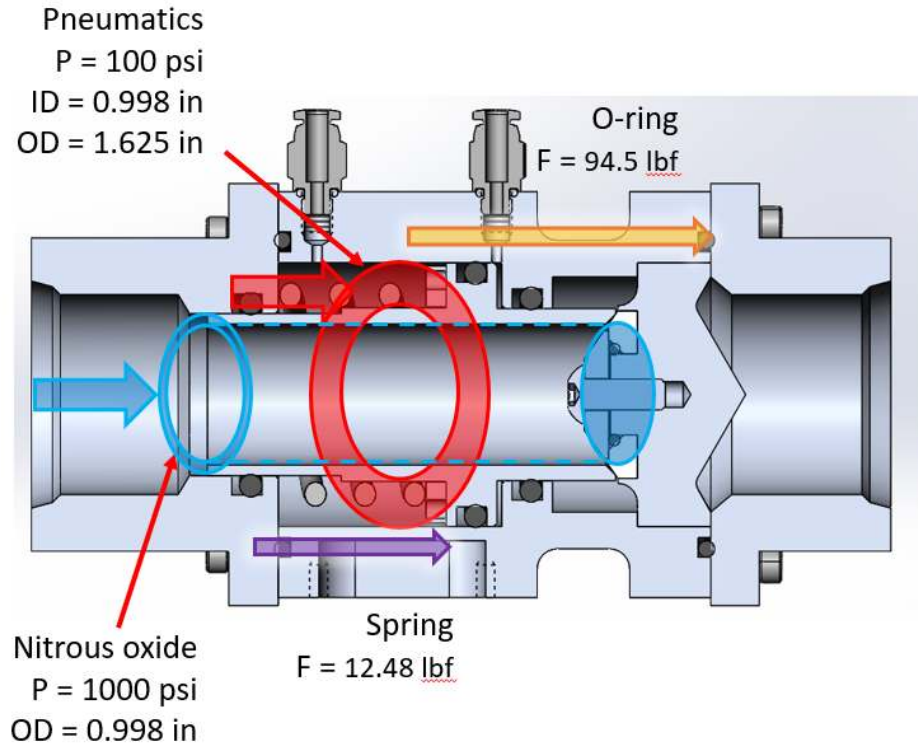


Figure 31: Areas and forces on outlet endcap, valve closed

The loads are summarized in Table 24 (positive force is in the *downstream* direction).

Table 24: Forces on outlet end cap when valve is closed

Location	Area (in ²)	Pressure (psi)	Force (lbf)
upstream sleeve+seat	0.7823	1000	782.26
Pneumatics	1.2917	100	129.17
O-ring compression	-	-	94.5
Spring compression	-	-	12.48
Net Force			1018.41

The load is 1018 lbf. Next, in the open case, the fluid pressure is acting on the inside of face of the endcap, to the extent of the face seal.

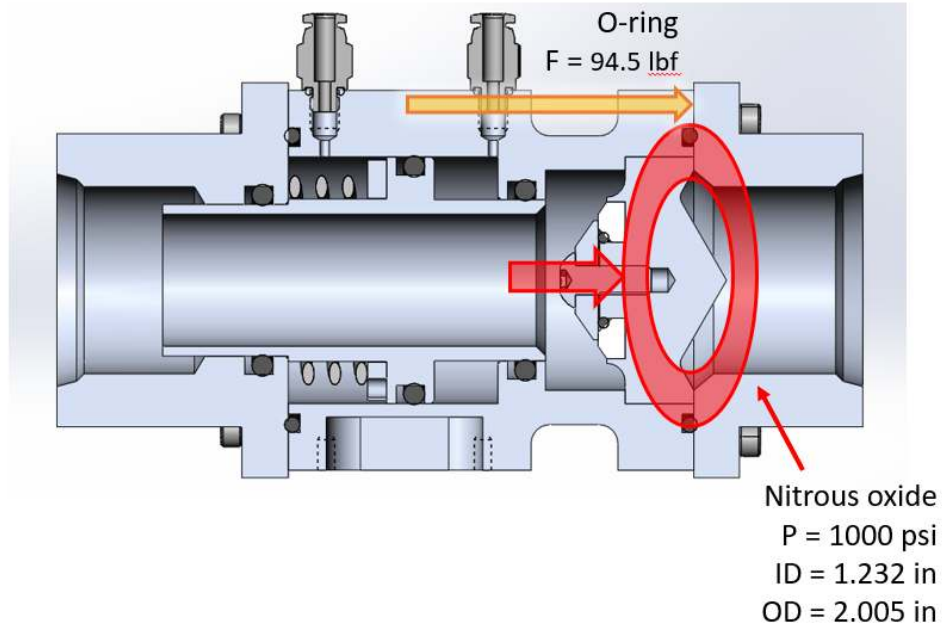


Figure 32: Areas and forces on outlet endcap, valve opened

The loads are summarized in Table 25 (positive force is in the downstream direction).

Table 25: Forces on outlet end cap when valve is opened

Location	Area (in ²)	Pressure (psi)	Force (lbf)
Endcap face	1.9652	1000	1965.22
O-ring compression	-	-	94.5
Net force			2059.72

Table 26: Required bolt tensile area calculations, outlet endcap, proof strength is for socket head screw per ASTM A574

Description	Value	Unit
Load	2059.72	lbf
Safety factor on yield	2	lbf
Proof strength	140000[10]	psi
Preload	75%	
Tensile area required	0.03923	in ²

The same bolts as on the inlet endcap will be used. This reduces the number of unique

parts in the assembly, which saves money ordering from unique parts and reduces errors in assembly.

Table 27: Size and number of bolts required for outlet endcap flange

Bolt size	Tensile Stress Area[11, p. 405] (in ²)	Bolts Required
#6 UNC	0.00909	4.3

Based on this calculation, a six-bolt pattern would be enough. However, since the inlet endcap also uses a six-bolt pattern, an eight-bolt pattern was used instead to differentiate the outlet end. This helps prevent the endcaps from being installed on the incorrect ends and the valve body from being installed in the reverse orientation.

Finally, the thread depth of is determined by calculating the required shear area of the female thread material such that the bolt will snap before the threads are torn out. In this case, if the bolts are over torqued or the flange overloaded, it is more desirable to have the bolt snap and be able to be extracted instead of tearing out the female threads and ruining the valve body. Both endcaps were machined out of 6061-T6 aluminum. A safety factor of 2 was used on the tear out failure mode, and the von Mises criterion was used to calculate the shear yield strength.

$$\text{Shear area} = \pi \cdot \text{diameter} \cdot \text{depth}$$

$$\frac{S_{sy}}{\text{SF}_{\text{tearout}}} = \frac{F}{A}$$

Table 28: Required thread depth calculations

Description	Value	Unit
SF _{tearout}	2	
Female thread major diameter	0.138[12]	in
S_y , Aluminum 6061-T6	40000[8]	psi
S_{sy} , von Mises	23094	psi
Bolt tensile strength	180000[10]	psi
Bolt tensile area	0.00909[11, p. 405]	in ²
Bolt tensile load	1636.2	lbf
Shear area required	0.1417	in ²
Required Thread Depth	0.327	in

The thread depth ended up being 0.45 in deep to allow for full thread profiles to be tapped to the calculated thread depth because the #6 tap in the student machine shop was not bottoming. The extra thread depth also accommodates variations in the thread length.

9 Position Feedback

Position feedback of the valve was identified as necessary to remotely ensure the valve was in a safe state (closed) prior to filling of the oxidizer tank on the rocket. This would involve being able to detect the physical location of the sleeve, instead of inferring the state of the valve based on the pneumatic pressure sent to its ports, since it is possible the sleeve fails to move when actuated if the valve gets cold. Some ideas explored included:

- 1) Sensing position of the sleeve using magnets with a Hall effect sensor
- 2) Sensing position of the sleeve using magnets with magnetic switches
- 3) Pressure sealed limit switch that would be mechanically actuated by the sleeve

The limit switch option was ruled out as it was too complex and impractical to implement, having sensors that would penetrate into the pressurized space of the piston actuator. The magnetic sensing options was preferred because the position of the sleeve can be sensed without physical contact, since the valve body is aluminum 6061. Out of the magnetic sensing options, the Hall-effect sensor option was preferable because it required fewer channels on the Rocket's control boards and was simpler to implement mechanically.

The sensor that was considered for this application was the TI DRV5055A1,

Since the magnet would have to be attached to the valve sleeve, it would have to be in an annular shape. To ensure the magnet did not slide along the length of the sleeve, it was decided to put it on the upstream side of the piston, where the load from the spring would always keep it in place. It was difficult to find a ring magnet that would meet the exact geometric requirements, so instead, a circular array of small magnets magnetized through the axis were used. Cylindrical 1/8" OD x 1/8" neodymium magnets McMaster P/N 5862K101 were chosen due to their strong magnetism. This particular size was selected for its low cost and because it would fit in the annular space of the piston. The magnets were held in place by pressing them into a waterjet piece of 1/8" PTFE, McMaster P/N 9266K84.

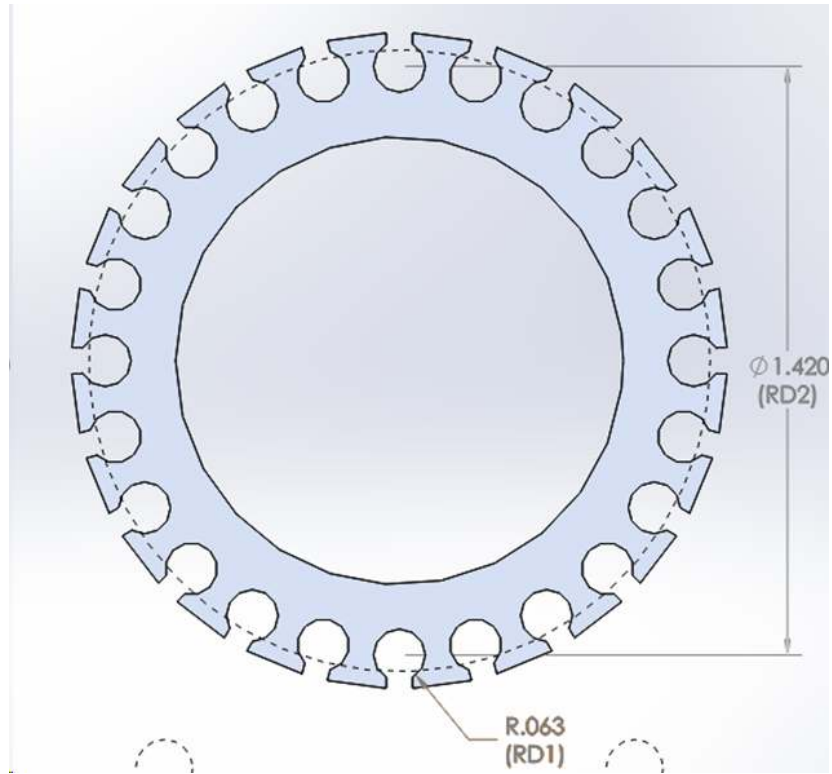


Figure 33: Magnet holder, SolidWorks screenshot

To maximize the strength of magnetic field, 24 magnets were arranged in a circular pattern with the diameter of the pattern increased to as far out as practical. Furthermore, to allow the sensor to be placed closer to the magnets, a pocket was also milled into the valve. This allowed detection of the magnetic field as close as 1 in to the center axis of the valve.

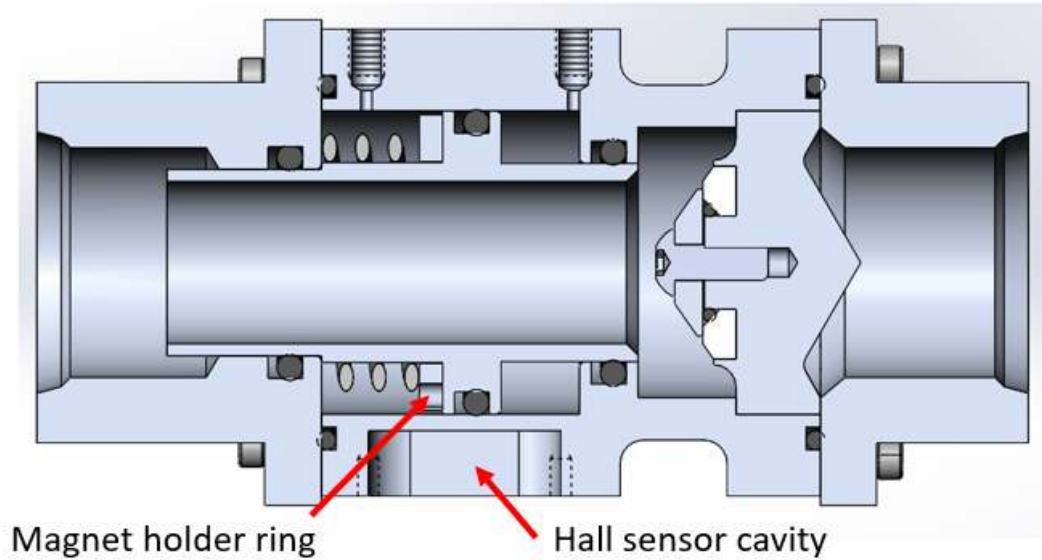


Figure 34: Position feedback features on valve

The performance of an earlier iteration of this circular array that used 16 magnets was simulated using the Python package Magpylib. Figure 35 shows the magnetic field strength in specific orientations depending on the distance from the center axis of the valve.

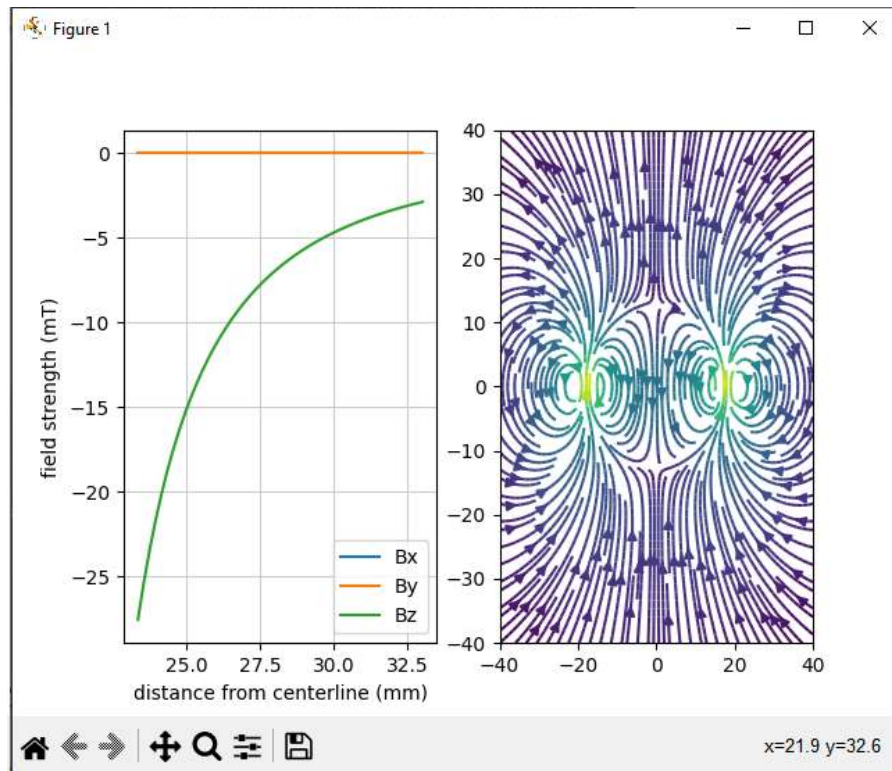


Figure 35: Magnetic field strength and field lines, 16-magnet array

The magnetic field reaches a strength ranging from 5 mT up to 15 mT when positioned in the plane of the magnets, depending on how deep inside the cavity the sensor is placed. Although this was determined to be acceptable, the final magnet count was increased to 24 to get a stronger signal on the magnets.

10 Testing

Machining of the valve was started in October 2022 (drawings are attached in the Appendix), with the first valve test article complete in mid-November ready for testing.



Figure 36: Valve first test article

Throughout the life cycle of the valve, component testing of the assembly and integrated testing as a part of the entire engine system were performed. The valve underwent actuation and hydrostatic proof tests first, and then was integrated on to the engine for cold flow and static fire tests. Finally, the valve was integrated into the rocket where it was part of a successful Wet Dress Rehearsal in preparation for transport and launch at SA Cap in June.

10.1 Actuation and position feedback testing

The assembled valve was first actuation tested to ensure the sleeve move as expected when pressurized with pneumatics. The valve was cycled by alternating pneumatic pressure between the open and close ports. Under ambient pressures, it took approximately 0.5 s to actuate the valve. This confirmed the pneumatic piston was moving as intended and allowed for progressing onto hydrostatic proof testing.

The position feedback system was also tested with a hall sensor measured by a multimeter, shown in Figure 37.



Figure 37: Compressed air being used to actuate the valve (bottom left), voltage reading displayed on multimeter.

The sensor output a voltage of about 2.2V closed and 4.5V open, which is a strong enough signal to use and validates the design of the magnetic sensing system.

10.2 Hydrostatic proof testing

After a prototype version of the valve was machined and assembled in late November, it was first hydrostatically proof tested using water up to 1.5x MAWP, for a pressure of 1500 psi to multiple times its expected operating duration. The P&ID for the test is shown in Figure 38.

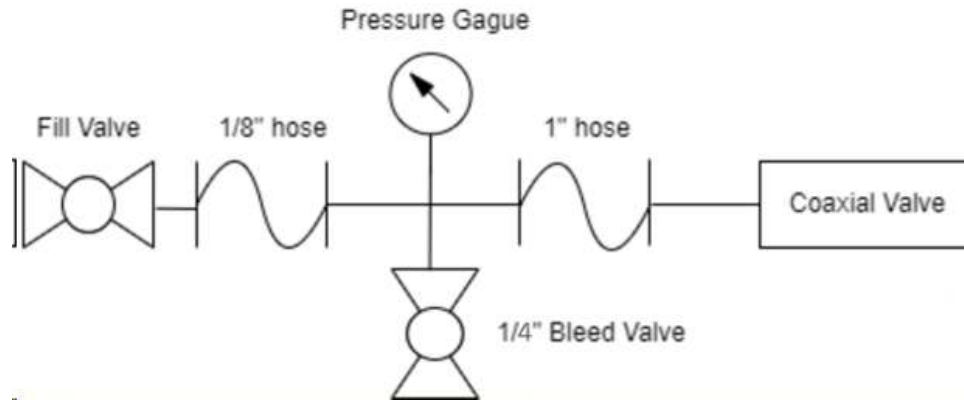


Figure 38: Valve hydrostatic testing P&ID

Water was first flowed through the assembly to remove trapped air, and then the bleed valve was closed. The coaxial valve was actuated closed using 100 psi compressed shop air. The pressure was then increased to the proof test pressure, at which point the fill valve was closed. The pressure decay was monitored using the gauge.

10.3 Valve seat radial deformation

The seat radial deformation issue was observed after the first round of hydrostatic testing of the valve. The first version of the valve seat holder had a flat surface that the valve seat rested on. After hydrostatic testing, it was found that secondary leakage through the seat retainer and into the inner diameter of the seat caused the seat to deform radially, as seen in the photo in Figure 39.



Figure 39: Seat radial deformation after hydrostatic testing

This effect happened gradually when exposed to pressures of 1000psi for prolonged periods of time (5min) and rapidly when experiencing hydrostatic proof test pressure of 1500psi. This specific failure mode was fixed by adding a lip feature to the valve seat holder, shown in Figure 40. This prevented the valve seat from extruding radially.

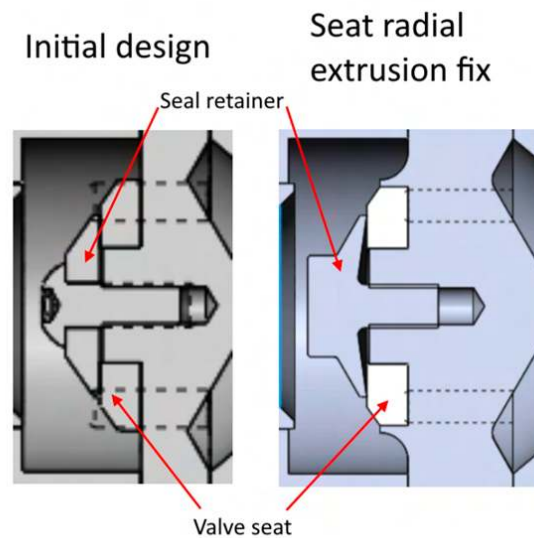


Figure 40: Revision of seat holder geometry to fix radial extrusion issue

10.4 Valve seat axial deformation

After Static Fire 7, the valve seat was found to have extruded out of the recess of the seat holder, bulging upwards around the seat retainer screw.



Figure 41: Comparison of valve seat before SF7 (left) and after SF7 (right)

This was after the valve experienced a prolonged period of time under pressure (1.5 h at up to about 800 psi), which was due to the extended amount of time it took during the Static Fire 7 test to warm the nitrous oxide propellant to the target firing pressure. The probable cause was again determined to be secondary leakage through the interface between the seat retainer and the valve seat. This time, the fluid leakage pushed on the back side of the valve seat, forcing it out of the seat recess. This leak path was sealed using an O-ring after modifying the valve seat geometry to accommodate a triangular O-ring gland. Figure 42 shows the addition of the secondary seal.

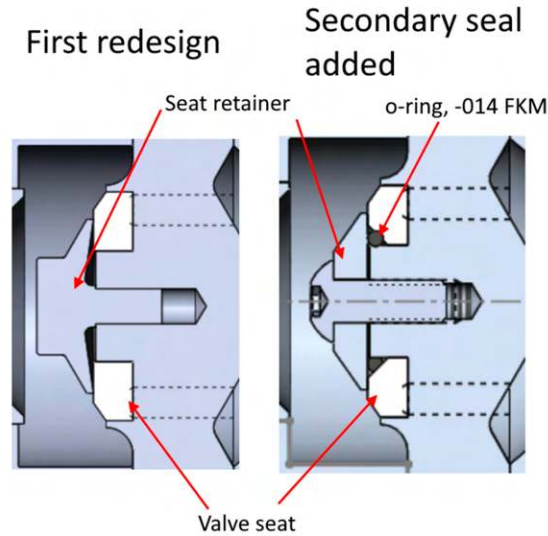


Figure 42: Revision of seat holder geometry to fix axial extrusion issue

Since this change was made, no further issues with seat deformation have been observed during subsequent testing where the valve experienced pressure for an extended amount of time.

10.5 Valve sticking issue

In the SF7 campaign, prior to the actual firing of the engine, a cold flow test using CO₂ was performed with the engine's injector elements without the rest of the combustion chamber attached. During this test the valve failed to actuate open when pneumatic pressure was at 100 psi. The pneumatic pressure was increased to 130 psi, which successfully actuated the valve for the engine firing. This issue was not observed during the SF8 campaign.

The likely root cause of this failure was determined to be very cold temperatures (below -10 degC), which caused the O-ring material to become stiffer. The stiffening of the O-rings increased the friction such that the actuator force was no longer sufficient to overcome the breakout force of the dynamic O-ring seals.

Furthermore, additional cause may have been moisture in the pneumatic systems getting into the dynamic seals of the sleeve and freezing, contributing to high friction. Moisture was discovered in the seals during disassembly of the valve after the engine test. The hypothesized mechanism is the moisture in the air condensing when the valve was brought outside from the University building to the test cell at Columbia Lake. Although it is possible that condensation after the valve was brought back indoors after the test could have caused the moisture that was seen during disassembly.

It was decided to use the hardware and design as is without changes because weather conditions at SA Cup are expected to be significantly warmer, at a mean average temperature of 28.8°C[13], making the possibility of valve sticking by cold and stiff O-rings remote.

10.6 Static Fire 7 and 8

The valve was used on January 29th, 2023, for the Static Fire 7 (SF7) engine test, and the flight article of the valve was tested on April 28 for the Static Fire 8 (SF8) engine test. Both were integrated hot fire test of the team's hybrid engine that also uses the flight tanks and feed system. SF7 featured major engine upgrades over the previous iteration of the engine that were aimed at improving combustion efficiency and thrust, and SF8 validated the flight engine design for the 2023 Rocket.

The valve was connected to the run tank through a series of 1" lines and was bolted to the injector bulkhead on top of the engine with a flanged adapter.

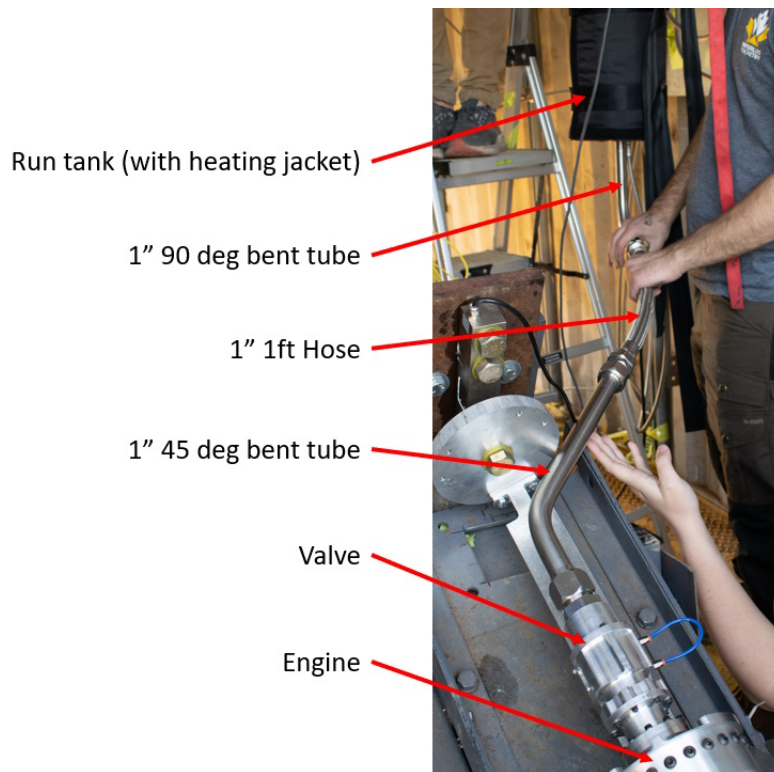


Figure 43: Feed line components at SF8

The valve operated nominally for both firings of the engine. The mass flow rate through the valve was measured using a load cell on the oxidizer run tank. The processed data from both tests are shown in the graphs in Figure 44 and Figure 45.

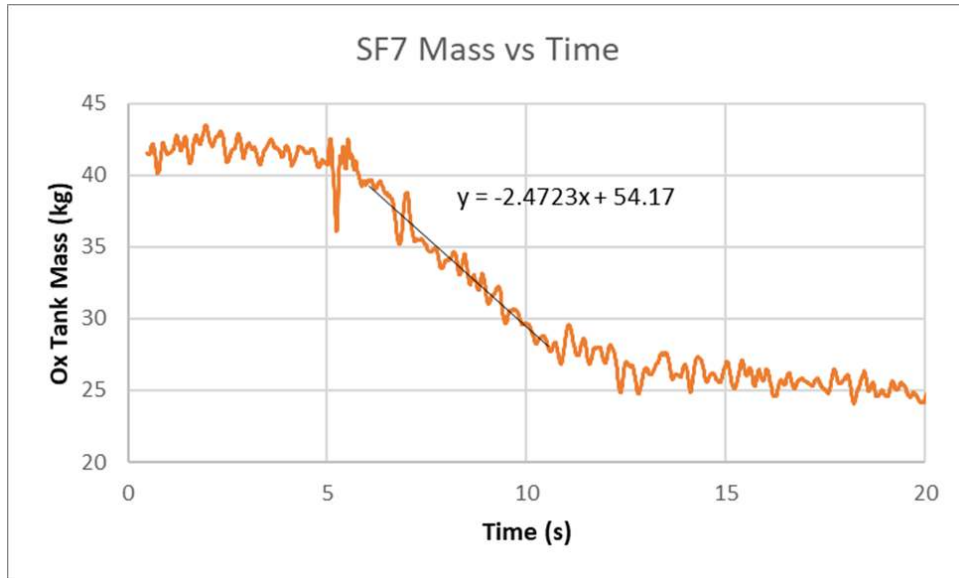


Figure 44: Mass of run tank vs time in SF7, linear fit through liquid portion of burn

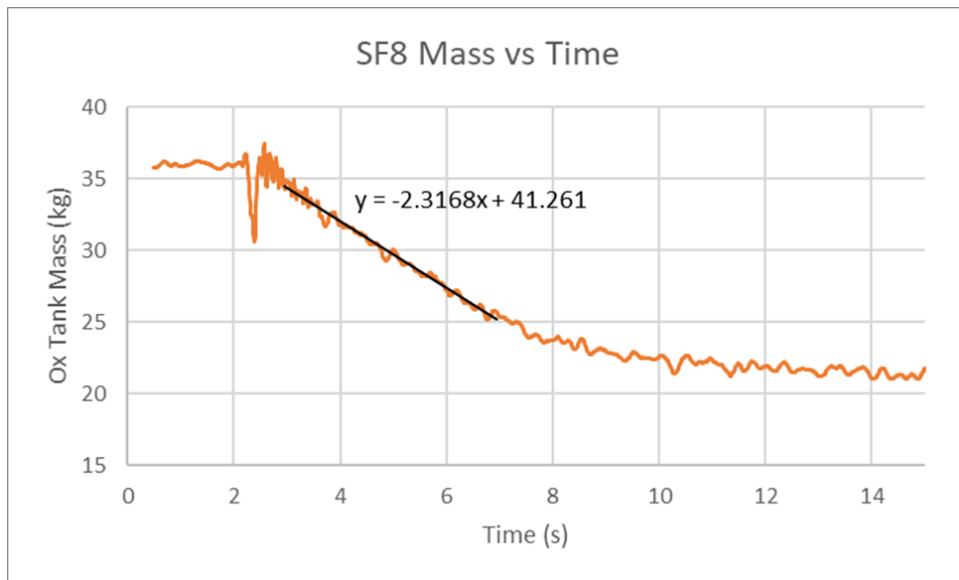


Figure 45: Mass of run tank vs time in SF8, linear fit through liquid portion of burn

When the injector valve first opens, liquid phase nitrous oxide is fed to the engine at firing pressure. As the tank empties, the ullage pressure decreases as the rate of vapor formation is not fast enough to keep up with the increasing ullage volume. Eventually, all liquid is expelled from the tank and the engine starts receiving the gas phase, which is indicated by the decrease in slope in the graphs. From the load cell data, the mass flow rate during the liquid phase of the engine burn, is about 2.5 kg/s and 2.3 kg/s for SF7 and SF8 respectively.

Although this mass flow rate was lower than the design target of 2.85 kg/s, flight dynamics

analysis by other Rocketry members determined that the engine performance from SF8 was sufficient to achieve the target altitude of 30,000 ft, so no further hardware changes to the valve were planned. Based on this conclusion, the design mass flow rate was likely too conservative on engine performance. Furthermore, the long length of feed line from the run tank to the valve in the SF7 and SF8 test setup may also have contributed to a lower flow rate. This length of flow path was dramatically reduced when the valve is integrated in the rocket, as shown in the Integration section.

SF8 was also the first test where the magnet-based position feedback system was used. This provided data on the actuation time of the valve, which was approximately 0.16 seconds from when the command is sent to the pneumatic solenoid to actuate the valve to when the valve reaches a fully open state, as shown in the Figure 46. The nitrous pressure in the valve at time of actuation was about 860 psi.

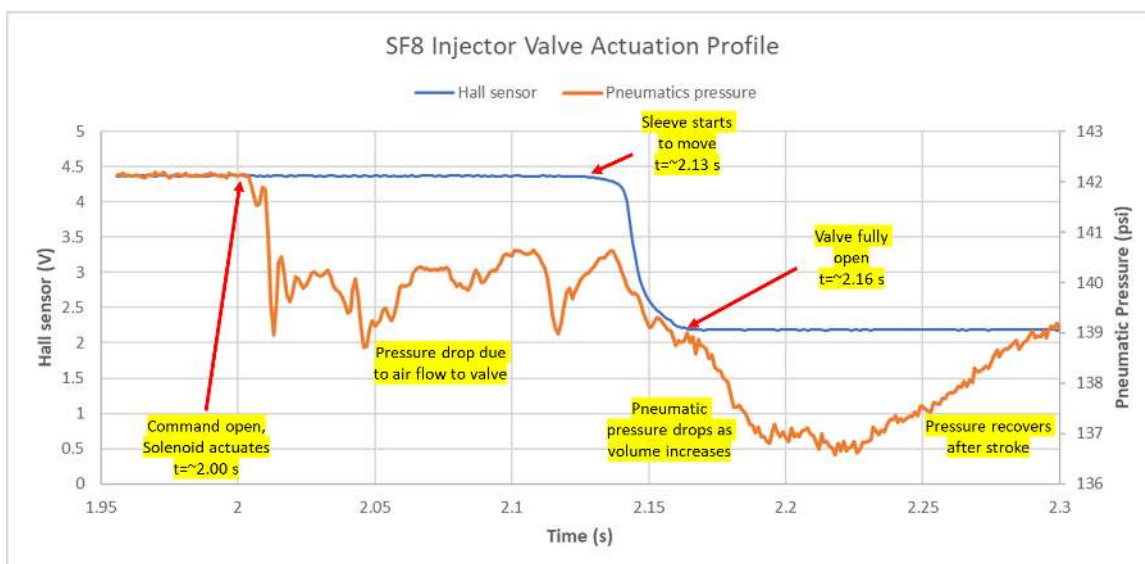


Figure 46: Actuation profile of valve during SF8, showing Hall-sensor and pneumatic pressure traces

With the performance of the valve deemed satisfactory, it proceeded to be integrated into the rocket. As part of the Rocket, it underwent Wet Dress Rehearsal testing to practice loading oxidizer on the rocket prior to launch at SA Cup. During the test, the valve performed as expected, providing good readings that confirmed the valve was in a closed position prior to loading propellant.

11 Integration

Flanged adapters were designed by other members of the Team to adapt the valve from the outlet of the oxidizer tank, which was the fill bulkhead, to the inlet of the injector bulkhead on top of the engine. This was a significant improvement in integration from the 2022 rocket integration, which used tapered threaded fittings (NPT). Compared to NPT, flanges require less insertion depth, and the tightness of the joint is not directly coupled to its angular alignment. The final assembled length is also predictable and much more consistent compared to a tapered thread, and did not have the risk of galling or stripping the thread if improperly assembled.

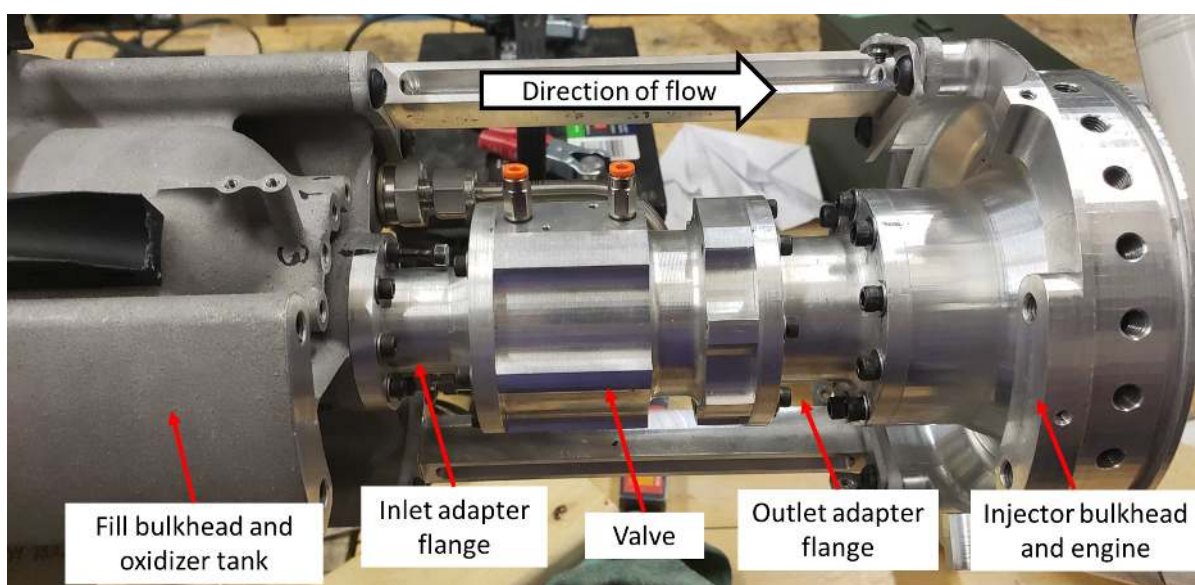


Figure 47: Injector section for the 2023 Rocket

The total mass of the assembly was measured to be 0.942 lb (about 427.3 g). This measurement included the valve body, sleeve, seat, seat holder, seat retaining hardware, spring, magnet holder and all magnets, Hall sensor wire leads and its mounting bracket, air fittings, and all fasteners. It does not include the mass of the adaptor flanges, air solenoid, and pneumatic tube. A photo of the components weighed is in the Appendix, Figure 52.

12 Results and Future Plans

The valve successfully actuated on June 21st, 2023, at the Spaceport America Cup, launching the 2023 Rocket, Leviathan of the Sky. The rocket achieved an apogee of 31,476 feet and reached a maximum speed of Mach 1.5. The increased engine performance due to the higher oxidizer flow rate and engine upgrades allowed the Rocket not only to hit the apogee target, but to produce such a clear exhaust plume that the launch rail could be seen from behind the exhaust.



Figure 48: Lift off of the 2023 Rocket at SA Cup

With the successful ascent of the Rocket, and a good burn from the engine, this valve project was deemed complete.

The Team is planning to build a rocket propelled by a liquid bipropellant engine in 2024, which will necessitate two separate propellant valves. These valves will be smaller than 1” due to the greater combustion efficiency of a liquid engine resulting in less mass flow rate required for each individual propellant. Part reduction on this design is also possible by combining the seat holder and the end cap into a single part.

Lastly, setup of additional component tests to better characterize the performance of future valves should be performed, particularly to determine the seat leakage rate, pressure drop (characterized with C_v or resistance C_dA), and actuation time. These parameters will be more important as the ignition sequence (order and timing that each propellant is introduced to the combustion chamber), and combustion characteristics will be more sensitive to upstream conditions for a liquid engine than a hybrid.

13 Conclusion

The need for a new injector valve assembly and its requirements were identified. The project was successfully executed from design, to manufacturing and testing, to integration and flight. The new coaxial valve addressed the issues that were present with the old injector valve assembly design. It allowed for higher oxidizer flow rate to the engine, easier assembly, reduced mass, and more compact integration in the rocket. The test campaign validated the overall architecture of the valve and found issues with its design that were able to be addressed or retired.

Furthermore, through this project, the Team gained experience with bolted flanges, custom pneumatic actuators, custom valve seats, metal to plastic sealing, and other technologies that will be applicable in future projects. The success of this project gives the Team greater confidence in pursuing custom valve designs for more advanced projects, such as a rocket powered by a liquid engine. This ability to design custom fluid system components and assemblies gives the Team greater flexibility with the performance and integration of its fluid systems, and reduces costs compared with being limited to available commercial options.

References

- [1] “What is a coaxial valve?” 2023. [Online]. Available: <https://www.coax-valve.com/what-is-a-coaxial-valve.html> (Accessed 2023-09-20).
- [2] O. Parker, “Parker o-ring handbook (ord 5700),” *Parker Seal Group, Lexington, KY*, 2021.
- [3] N. Christopher, N. H. Wong, and S. Dalglish, 2018.
- [4] “36 series stainless steel 2-way full port ball valve, user’s manual,” 2023. [Online]. Available: https://assuredautomation.com/literature/IM/AA_User_Manual_36.pdf (Accessed 2023-09-20).
- [5] “C series dual rack-n-pinion pneumatic valve actuators,” 2023. [Online]. Available: https://assuredautomation.com/literature/C_pneumatic_actuators.pdf (Accessed 2023-09-20).
- [6] “36 series stainless steel ball valves,” 2023. [Online]. Available: <https://assuredautomation.com/36/> (Accessed 2023-09-20).
- [7] “Smc vqz1121-6l1-n1 valve, body ported (dc), vqz1000 valve, sol 4/5-port,” 2023. [Online]. Available: <https://www.smc-pneumatics.com/VQZ1121-6L1-N1.html> (Accessed 2023-09-20).
- [8] “Aluminum 6061-t6; 6061-t651,” 2023. [Online]. Available: <https://asm.matweb.com/search/SpecificMaterial.asp?bassnum=ma6061t6> (Accessed 2023-09-20).
- [9] R. Lynch, “Two/2/way, latching, dc solenoid control valve final report, 1 mar.-28 dec. 1965,” Tech. Rep., 1966.
- [10] “Mechanical properties of externally threaded fasteners,” 1 2016. [Online]. Available: <https://www.fastenal.com/content/feds/pdf/2016/01/Mechanical%20Properties%20Inch%20Fasteners%20Jan%202016.pdf> (Accessed 2023-09-20).
- [11]
- [12] “Ansi internal screw threads size and tolerances table chart,” 2023. [Online]. Available: https://www.engineersedge.com/thread_strength/internal_screw_threads_chart.htm (Accessed 2023-09-20).
- [13] “Climate weather.gov > el paso, tx > climate,” 2023. [Online]. Available: <https://www.weather.gov/wrh/Climate?wfo=epz> (Accessed 2023-09-20).

Appendix

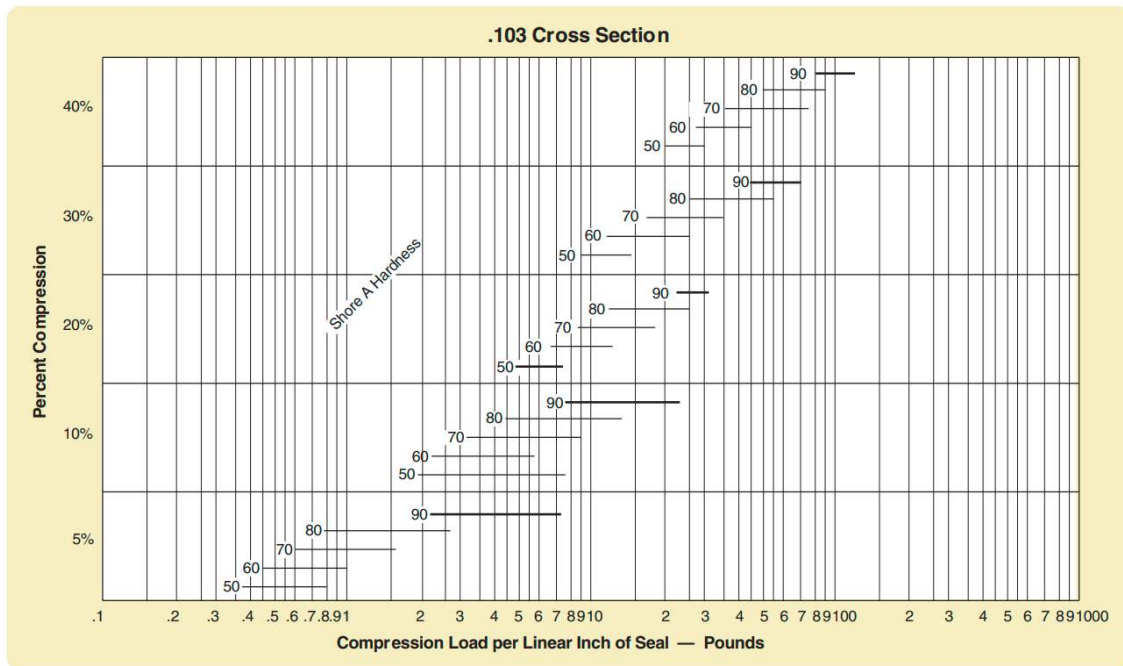


Figure 2-5: .103 Cross Section

Figure 49: Compression load per linear inch of seal, .103 width O-ring[2, p. 2-11]

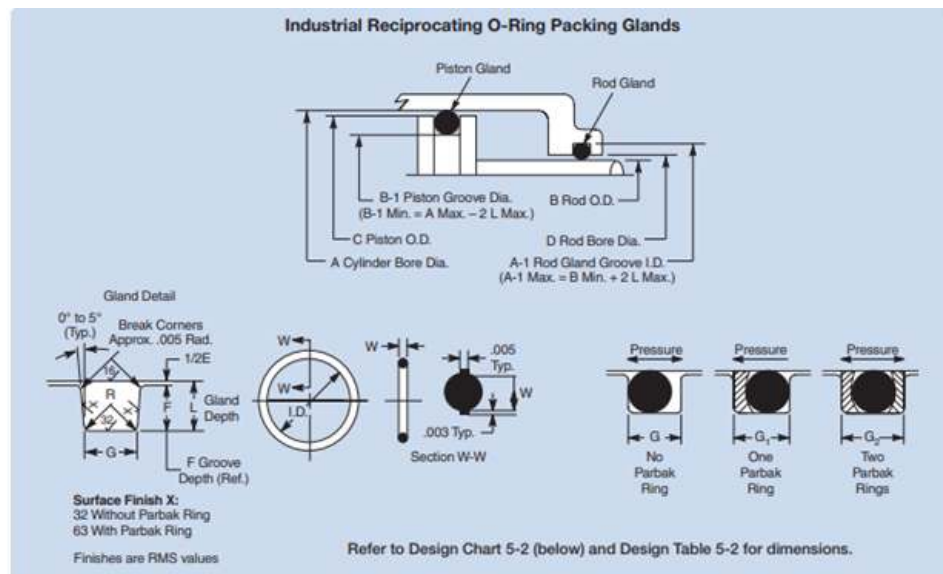


Figure 50: Industrial reciprocating O-ring gland diagram[2, p. 5-31]

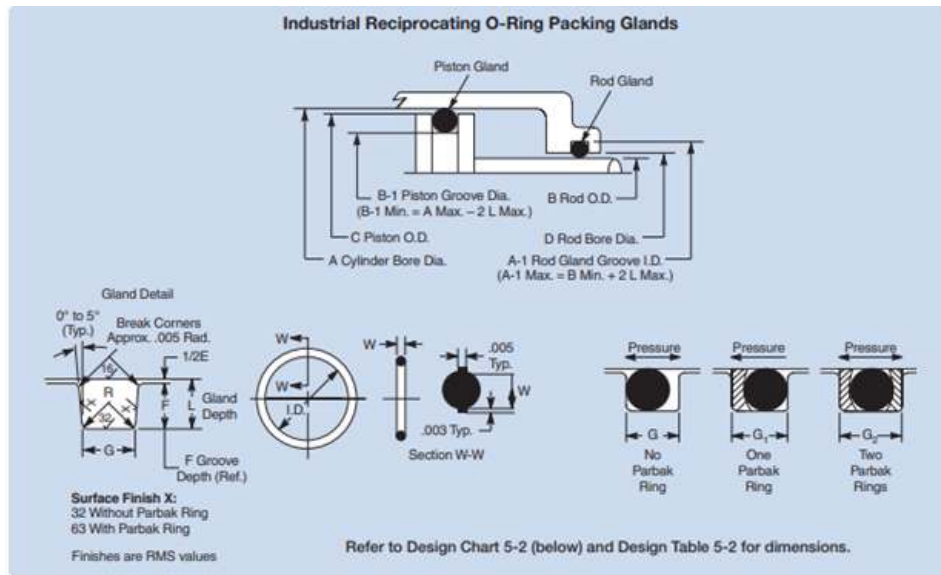


Figure 51: Gland Dimensions for Industrial Reciprocating O-Ring Seals, 103.5 Bar (1500 psi) Max[2, p. 5-32]



Figure 52: Flight valve components being weighed

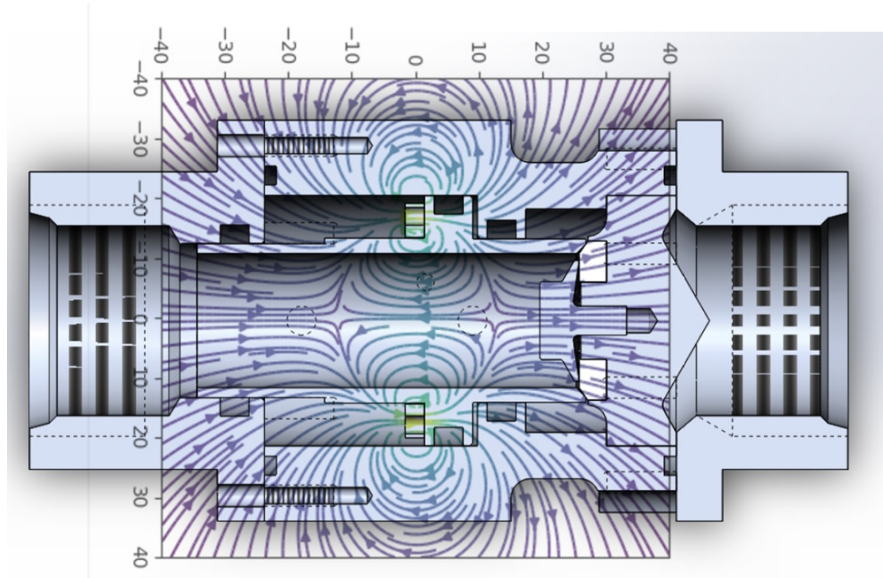


Figure 53: Magnetic field of the magnets overlayed on the cross section of the valve

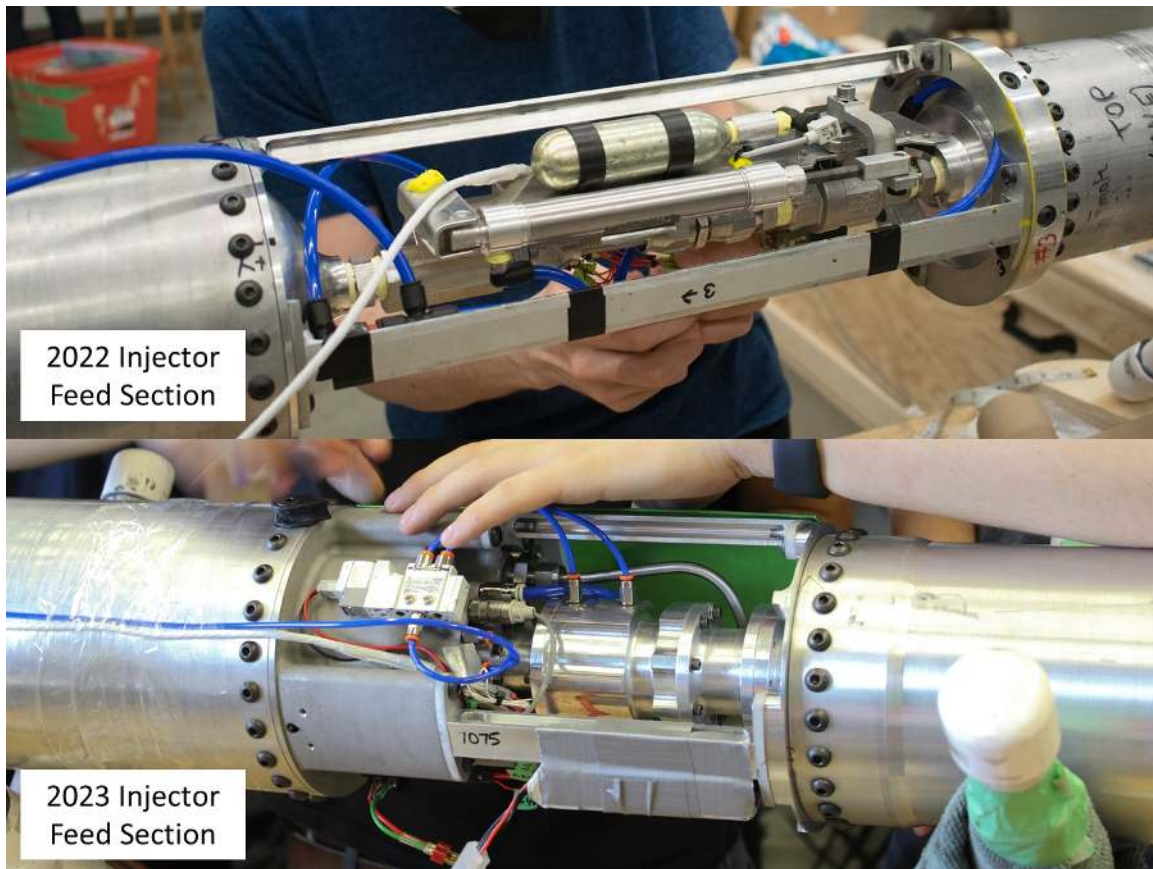


Figure 54: Comparison between the 2022 Rocket injector section (top) and the 2023 Rocket injector section (bottom)

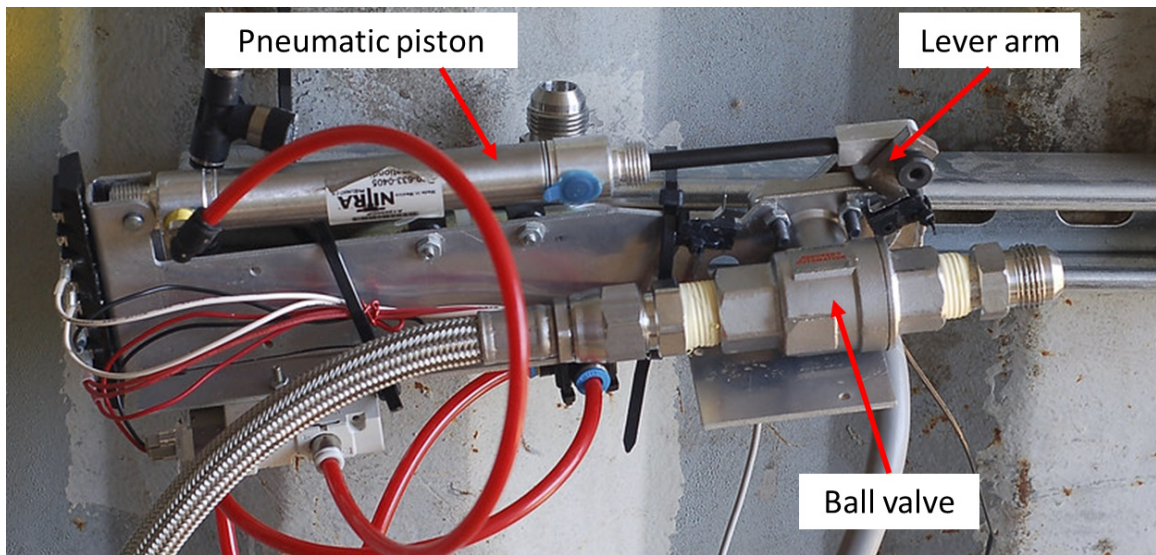


Figure 55: 2022 injector valve assembly, "FAST"

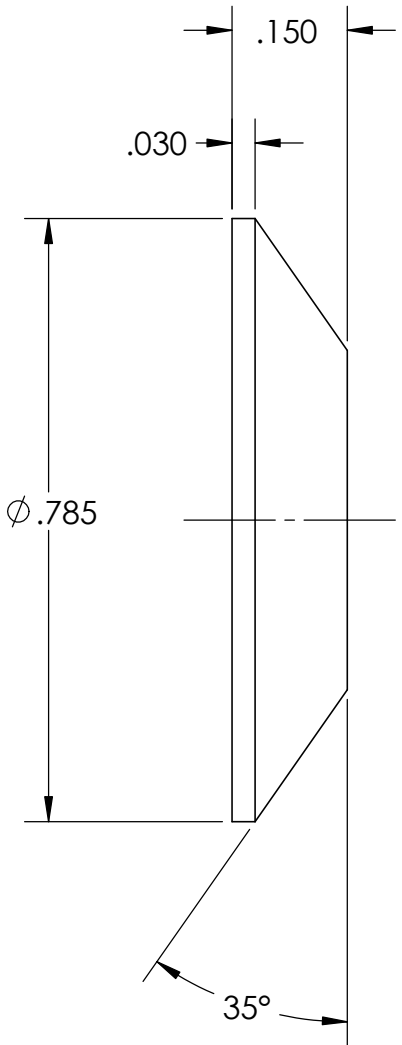
Table 29: Purchased components, flight valve

Part number	Description	Vendor	Qty	UOM
TI DRV5055A1QLPG	Hall Effect Sensor Single Axis TO-92-3	Digikey	1	EA
MS18-10N	NITRA pneumatic push-to-connect fitting, male, straight, nickel-plated brass body, nickel-plated brass threads, 1/8in tube to #10-32 threaded connection(s)	Automation Direct	2	EA
92949A265	18-8 Stainless Steel Button Head Hex Drive Screw 10-32 Thread Size, 1/2" Long	McMaster-Carr	1	EA
9464K35	Chemical-Resistant Viton® Fluoroelastomer O-Ring 1/8 Fractional Width, Dash Number 214	McMaster-Carr	1	EA
9464K36	Chemical-Resistant Viton® Fluoroelastomer O-Ring 1/8 Fractional Width, Dash Number 215	McMaster-Carr	1	EA
91251A151	Black-Oxide Alloy Steel Socket Head Screw 6-32 Thread Size, 3/4" Long	McMaster-Carr	14	EA
92141A008	18-8 Stainless Steel Washer for Number 6 Screw Size, 0.156" ID, 0.312" OD	McMaster-Carr	14	EA
9464K133	Chemical-Resistant Viton® Fluoroelastomer O-Ring 3/32 Fractional Width, Dash Number 133	McMaster-Carr	2	EA
9464K42	Chemical-Resistant Viton® Fluoroelastomer O-Ring 1/8 Fractional Width, Dash Number 220	McMaster-Carr	1	EA
5862K101	Neodymium Magnet Magnetized Through Thickness, 1/8" Thick, 1/8" OD	McMaster-Carr	24	EA
9663K42	Corrosion-Resistant Compression Spring Stock 20" Long, 1.375" OD, 1.079" ID	McMaster-Carr	1.027	IN

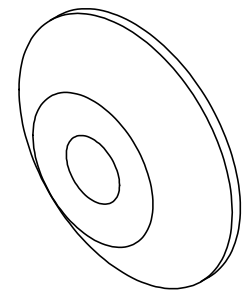
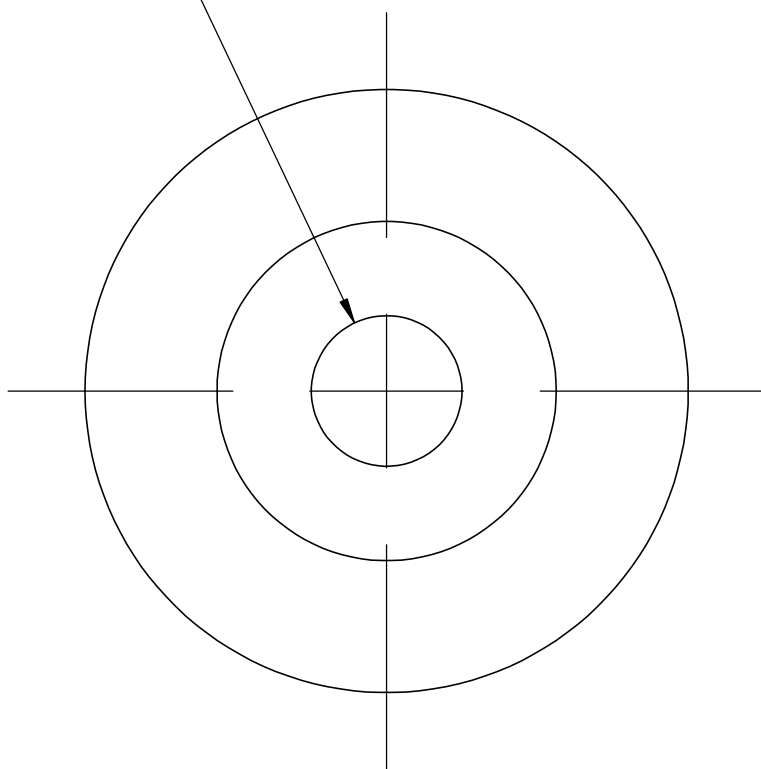
Table 30: Fabricated parts, flight valve

Component	Stock	Method of manufacture	Qty	UOM
Body	Aluminum round, 7075-T6, 3" dia.	Machined in-house	1	EA
Sleeve	Aluminum round, 6061-T6, 2" dia.	Machined in-house	1	EA
Seat holder	Aluminum round, 6061-T6, 2" dia.	Machined in-house	1	EA
Seat retainer	Aluminum round, 6061-T6, 1" dia.	Machined in-house	1	EA
Seat	25% glass-filled PTFE Rod, 1-1/8" dia.	Machined in-house	1	EA
Hall-sensor mount plate	PLA	3d-printed in-house	1	EA
Magnet holder ring	PTFE sheet, 1/8" thick	Waterjet	1	EA

REVISION HISTORY			
REV	DESCRIPTION	DATE	AUTHOR
00	INITIAL RELEASE	2022-10-04	M. ZHOU



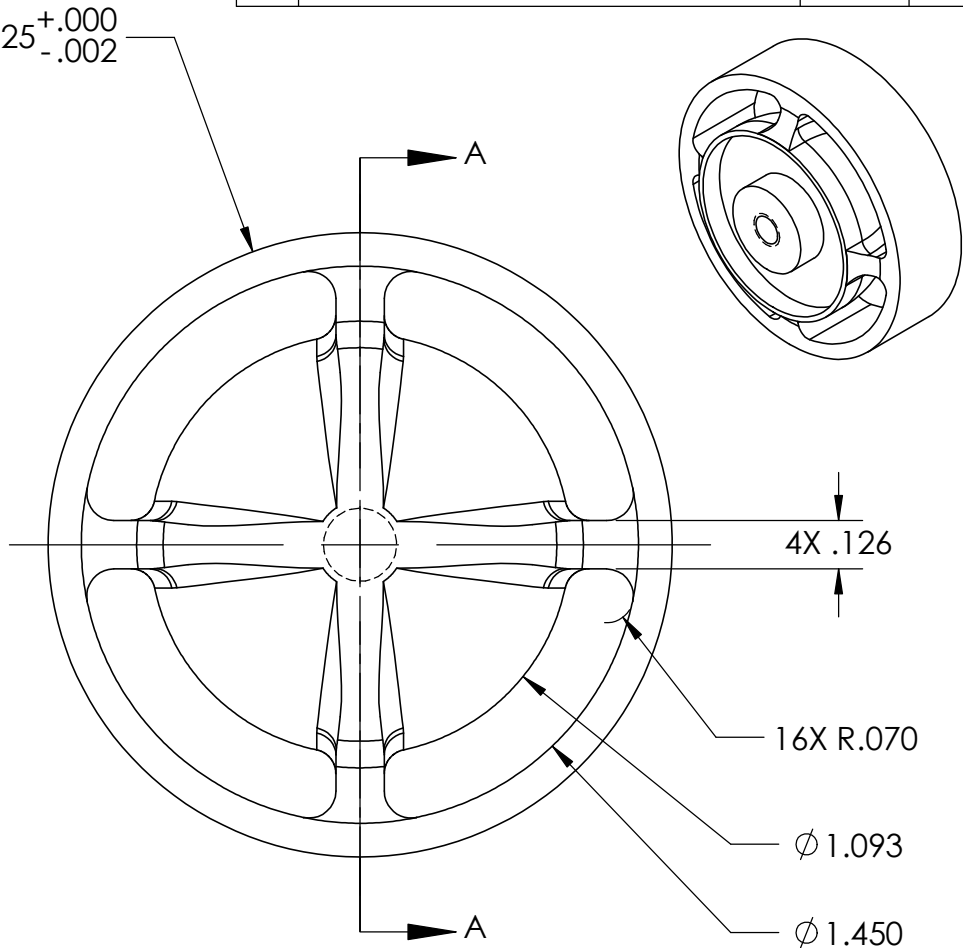
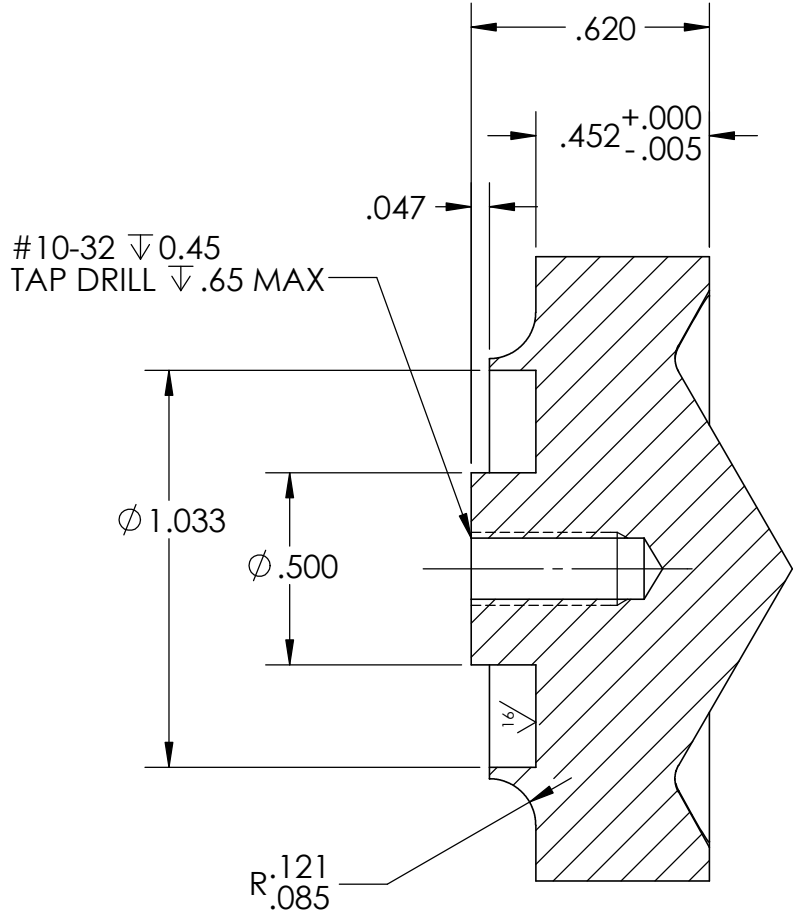
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UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES REMOVE BURRS AND SHARP EDGES TOLERANCES: LINEAR: 2 DECIMALS: $\pm 0.010"$ 3 DECIMALS: $\pm 0.005"$ ANGULAR: $\pm 1^\circ$		
	NAME	DATE
DRAWN	M. ZHOU	2022-10-04
MATERIAL:		
AL 6061-T6		
FINISH:		
AS MACHINED		
WEIGHT:	0.00	LBS

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DESCRIPTION:		
SEAL RETAINER		
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REVISION HISTORY			
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REMOVE BURRS AND SHARP EDGES
TOLERANCES:
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3 DECIMALS: $\pm 0.005"$
ANGULAR: $\pm 1^\circ$



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MATERIAL:		
6061-T6		
FINISH:		
AS MACHINED		
WEIGHT:	0.02	LBS

DESCRIPTION:		
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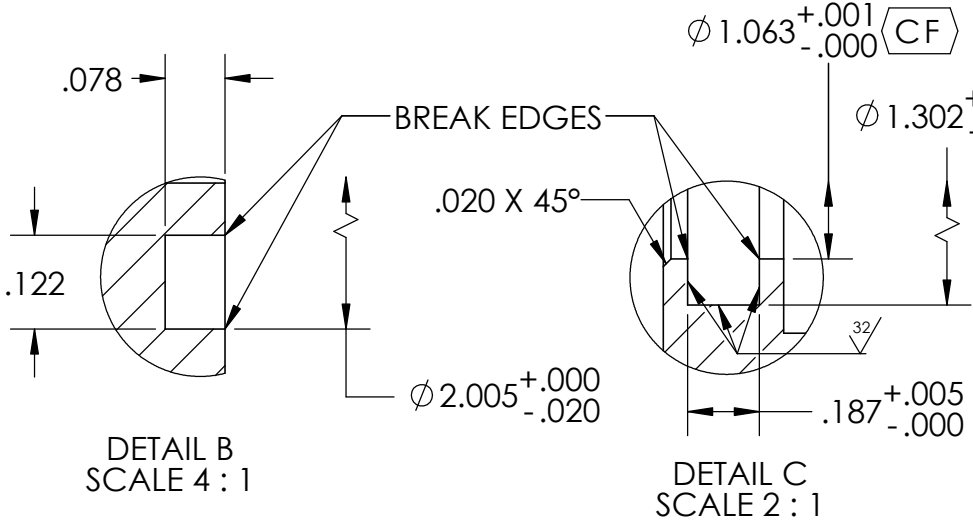
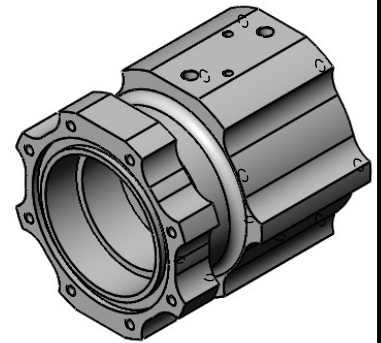
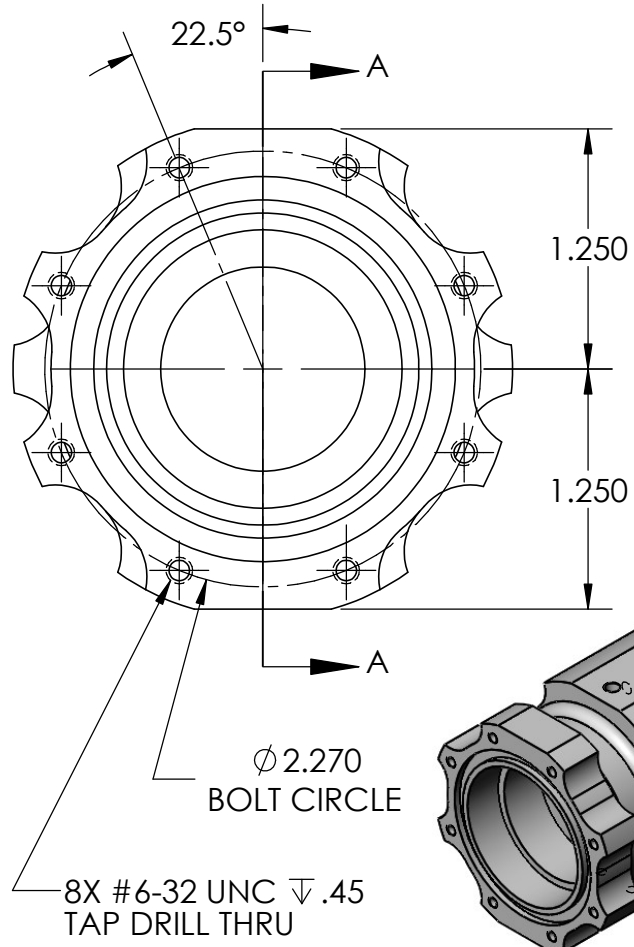
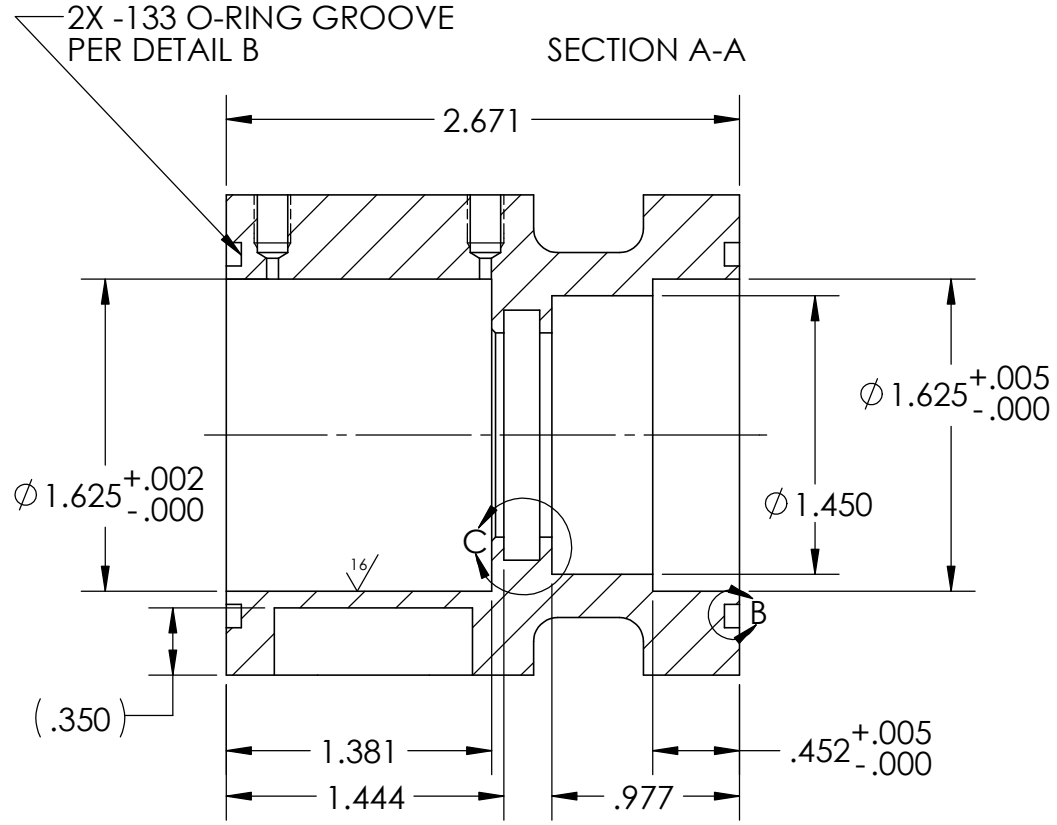
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NAME	DATE	
DRAWN M. ZHOU	2023-03-09	
MATERIAL:	AL7075-T6	
FINISH:	AS MACHINED	
WEIGHT:	0.59	LBS

 WATERLOO ROCKETRY		
DESCRIPTION: VALVE BODY, FLIGHT		
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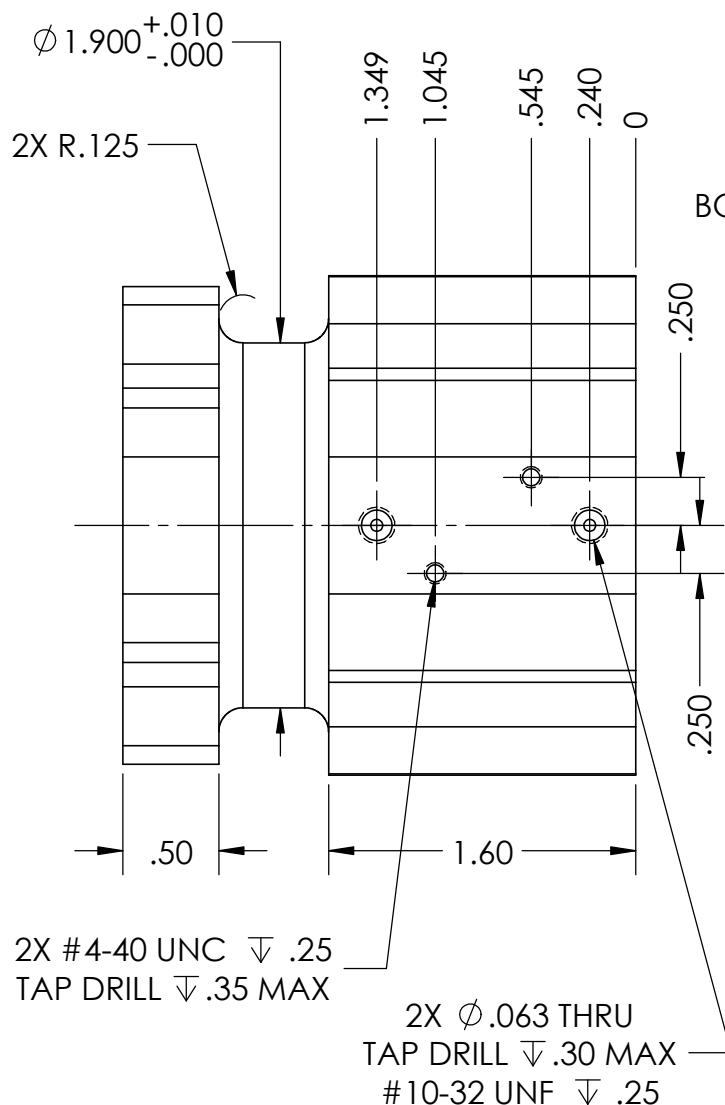
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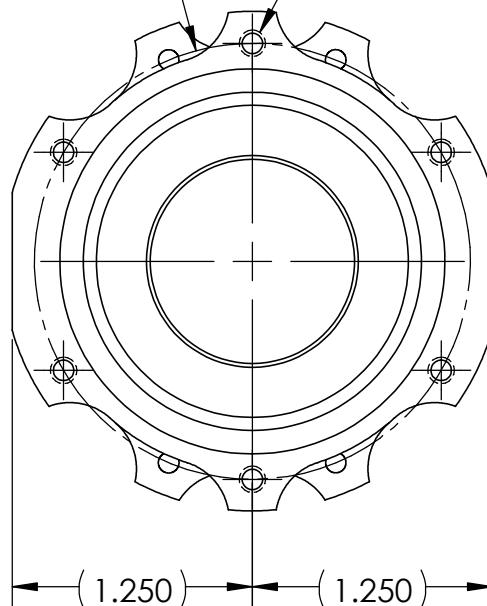
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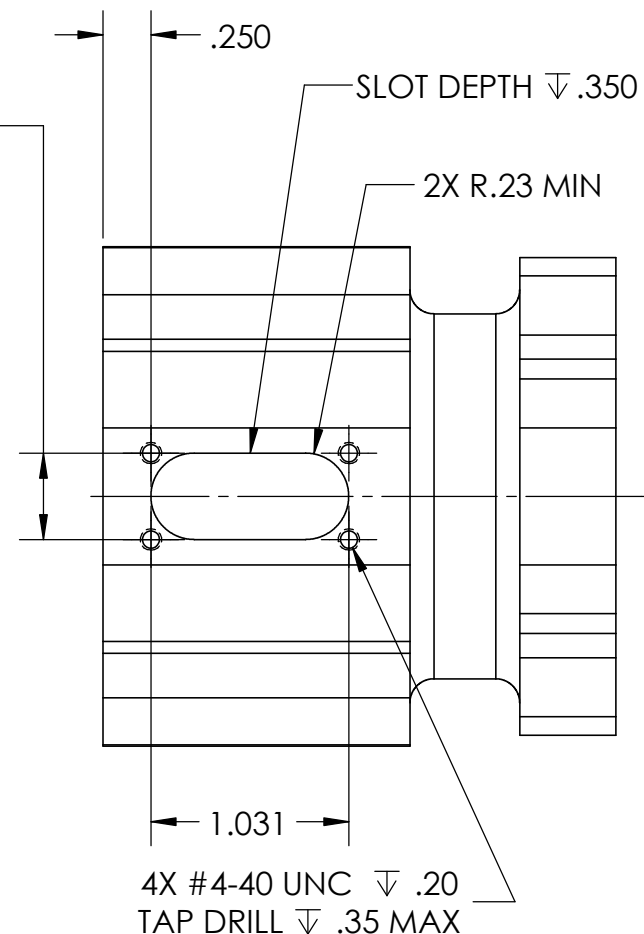


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TAP DRILL ∇ 1.0 MAX

$\phi 2.270$
BOLT CIRCLE



.450



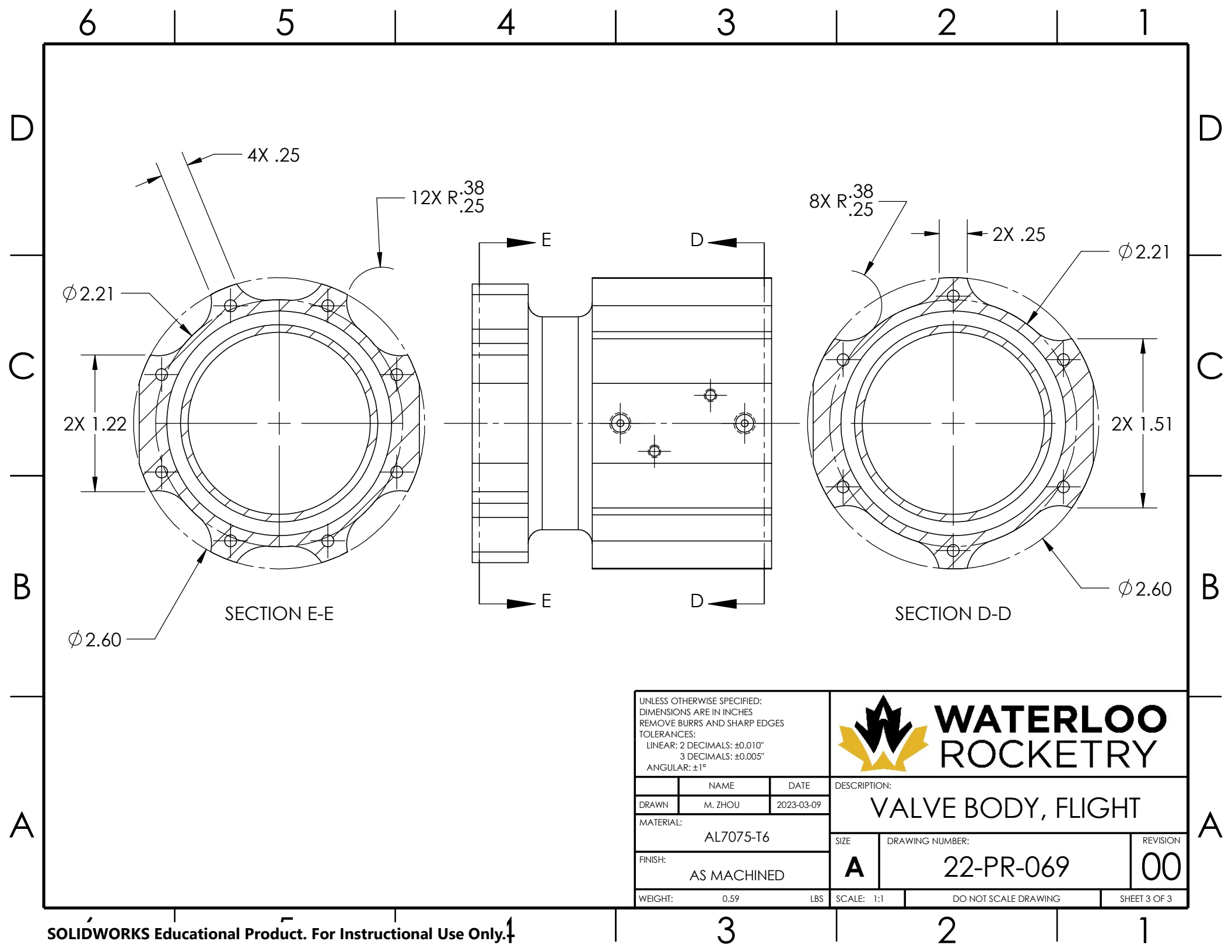
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3 DECIMALS: $\pm 0.005"$
ANGULAR: $\pm 1^\circ$



**WATERLOO
ROCKETRY**

	NAME	DATE
DRAWN	M. ZHOU	2023-03-09
MATERIAL:	AL7075-T6	
FINISH:	AS MACHINED	
WEIGHT:	0.59	LBS

DESCRIPTION: VALVE BODY, FLIGHT		
SIZE A	DRAWING NUMBER: 22-PR-069	REVISION 00
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REVISION HISTORY			
REV	DESCRIPTION	DATE	AUTHOR
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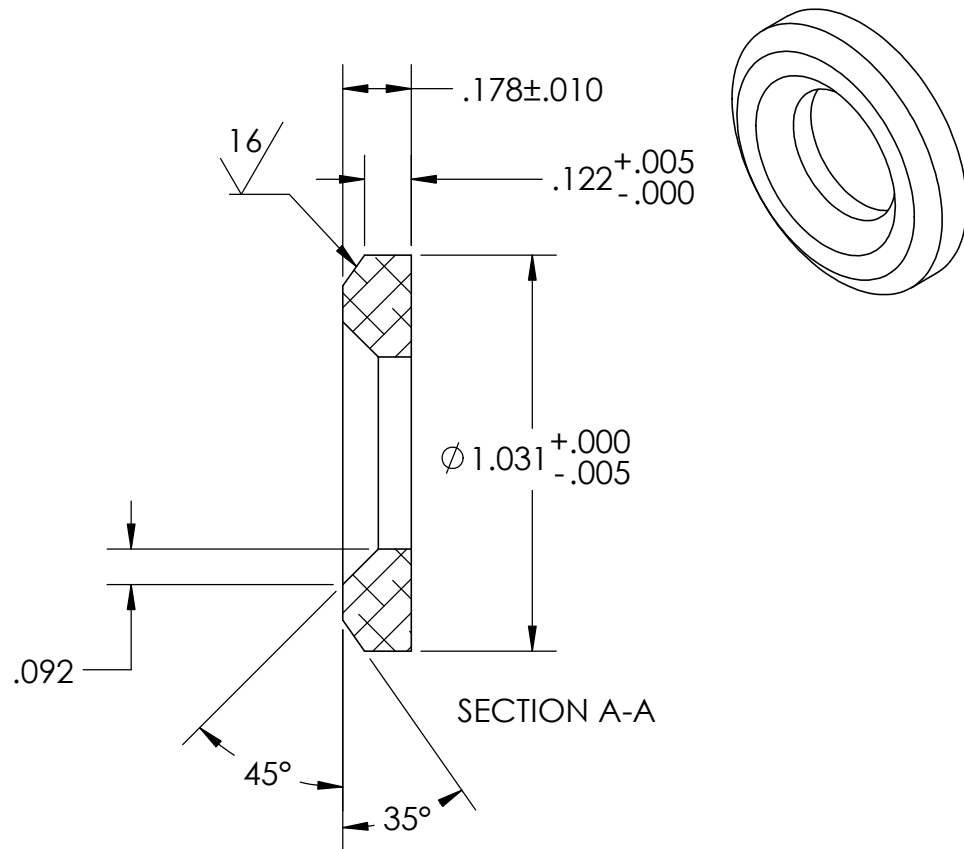
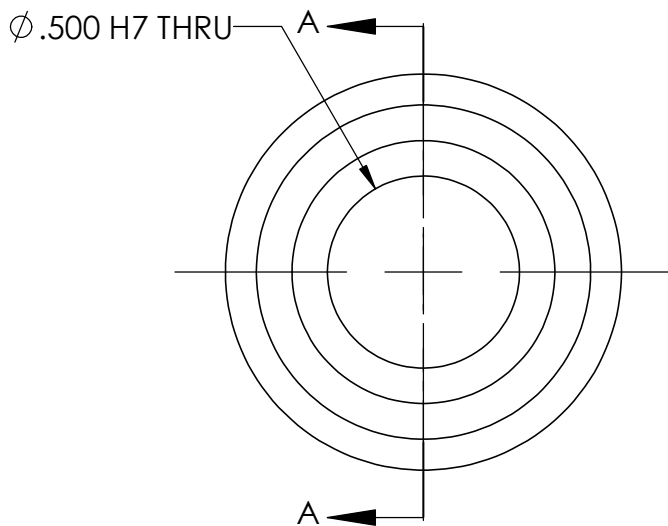
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REVISION HISTORY			
REV	DESCRIPTION	DATE	AUTHOR
00	INITIAL RELEASE	2022-10-04	M. ZHOU
01	CHANGE VALVE SEAT GEOMETRY FOR NEW SEAT HOLDER	2023-01-13	M. ZHOU
02	ADDED SECONDARY SEAL GLAND	2023-04-10	M. ZHOU



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES REMOVE BURRS AND SHARP EDGES TOLERANCES: LINEAR: 2 DECIMALS: ±0.010" 3 DECIMALS: ±0.005" ANGULAR: ±1°			 WATERLOO ROCKETRY		
			DESCRIPTION: COAXIAL VALVE SEAL		
DRAWN	NAME M. ZHOU	DATE 2023-04-10	SIZE A		
MATERIAL: 25% GF PTFE					
FINISH: AS MACHINED			DRAWING NUMBER: 22-PR-048		REVISION 02
WEIGHT: 0.01		LBS	SCALE: 2:1		DO NOT SCALE DRAWING
SHEET 1 OF 1					